CTDMS

Central Transport & Dispatch Management System

Version 1.6 - Master User Manual

Demonstration Edition

One master workbook · Depots (several in one city) · Customer types and codes · Invoice Recognition Code · Local autos per trip / per box · Keyboard-first operation

Prepared for Tulsyan | Application version 1.6.0 | October 2026

Every master record, invoice and gatepass in this document is a DEMONSTRATION ENTRY (codes DM-..., names ending (Demo)), created on a clean demonstration database that is separate from live data.

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1. Version 1.6 - Changes Completed

What management asked for after the version 1.5 review, and where it now is in the application.

#	Request	What was done	Where in CTDMS
1, 11	One-time master upload	One workbook (Instructions + 11 sheets), validated row by row, preview with accepted / rejected counts, import of valid rows only, history and Excel error report; existing codes never duplicated; later additions one by one	Master Settings › Initial Master Upload
2	CFA → Depot	Every screen, print, report, export and role says Depot (Depot Master, Depot Manager, Depot-wise report); no business-type split	everywhere
3	Several depots in one city	Depot code (unique), name, locality, city, state, PIN, address, companies handled, source ERP, active	Master Settings › Depots
4	Customer master and types	Customer types are a master (not fixed in the program); customers have type, source ERP and its customer code, depot and company mapping	Master Settings › Customers / Customer Types
5-7	Customer / type codes, recognition code, start + end of the number	Short recognition codes on company, customer type, customer and depot. Code = prefix (company, type, customer, depot) + numeric part of the ERP number, pattern configurable. No invoice-range feature	Master Settings › Invoice Recognition Code
6	ERP number not replaced	The original ERP invoice number is stored and shown exactly as received; the code is a separate field	register, invoice, slip, reports, API
8	Recognition logic	Company, depot and customer type recognised from the masters, ERP mappings and existing invoice markings (Invoice Code Mappings); a missing code gives a clear reason	imports, QR scan, typing, API
9	Coding as a provision	Switch on / off, pattern changeable, dispatch never depends on it, regenerate codes after master changes	Invoice Recognition Code page
10	No product / SKU work	Item-wise exports still accepted, lines not stored or shown; SKU report hidden; can be switched on later	System Settings › features.sku_lines (off)
12-16	Local autos per trip / per box	Local Auto Transporters with allowed methods (per trip, per box, both), usual trip charge, rate per box; per-box rates by location / company / depot / date; one trip amount shared by boxes, never multiplied	Master Settings › Local Auto Transporters / Box Rates; gatepass
17	Simple daily flow	Invoice in → recognised → boxes → carrier (long distance unchanged, or local auto) → slip → print → gate-out → MIS	Dispatch desk
18	Master Settings	All masters in one menu; active / inactive instead of deleting	Master Settings menu
19	Reports and search	Filters for recognition code, depot, city, customer type, local-auto method; City-wise, Depot-wise, Local Auto Trips; ERP number and code in separate columns	Reports, Invoice Register
20-21	Demo data and video	New v1.6 demonstration workbook and invoices; the guide video re-recorded in the same style	this manual, the video
22	Keyboard first, mouse optional (MARG style)	Same function keys on every screen (F1 help, F2 new, F3 go to, F4 search, F6 save, F7 edit, F8 gatepass / print, F9 complete dispatch, F10 reports, Esc back); Enter moves field to field and then to the main button; choice boxes find a company, depot, customer, customer type, area or transporter by code or name; arrows + Enter in every list; return to the same row; clear yellow focus; key bar at the bottom of every screen; delete / void / cancel ask first with No pre-selected	every screen - F1 shows the keys

2. Summary for top management

How version 1.6 works, end to end (243 words)

One clean start. On the first day the master data from MARG, SAP, CBO and Intrect is uploaded once in one workbook. CTDMS checks every row, shows what is accepted and what is wrong, and creates only valid records. New companies, depots, customers or transporters are added later one by one. **Depots, not CFA.** A depot is the godown goods leave from; one city can have several, such as Guma and Kamal Vihar in Raipur. Reports work by city and by depot. **Codes that explain an invoice.** Company, customer type, customer and depot each have a short code. Together with the number part of the ERP invoice they form the Invoice Recognition Code, for example ABC-RET-C025-GUMA-001245: ABC Pharma, retailer, customer C025, Guma depot, invoice 001245. The original ERP number is never changed. The code is a provision: it can be switched off or changed, and dispatch never depends on it. **Local autos charged correctly.** A per-trip auto is owed one amount for the trip however many invoices it carries; the amount is only shared over the invoices for the MIS. A per-box auto is boxes times its rate. Long-distance transporters work as before. **Keyboard first.** Staff used to MARG work with codes, Enter, arrows and function keys; the mouse is optional. **Result:** one consistent master list, invoices recognised automatically, simple daily dispatch, and transportation expense that adds up exactly - prepared for using the same structure across CFA, distribution and super-stockist work later.

3. Read this first - demonstration entries

DEMONSTRATION The demonstration started from a completely empty database. Only the administrator login was created on the command line. Everything else came from the demonstration master workbook or was entered through the screens, and the video shows it.

Kind	How it is marked	Example
Master codes	start with **DM-**	DM-ABC, DM-GUMA, DM-C025
Names	end with **(Demo)**	ABC Pharma Ltd (Demo)
Recognition codes	short codes as management asked (they are what appears in the Invoice Recognition Code)	ABC, GUMA, C025
Files	start with **DEMO_v1.6_**	DEMO_v1.6_Master_Workbook.xlsx
Screens and video	orange banner 'DEMONSTRATION DATA - NOT REAL'	every video frame

The demonstration has its own database (**v1.6/demo_database/ctdms_v16_demo.db**, started with **start_demonstration_v16.bat** on port 8004). Live data is never mixed with it. To go live, start from an empty database and upload your own master workbook.

Delivered	File
This manual	v1.6/CTDMS_v1.6_Master_User_Manual_Demonstration_Edition.pdf
Guide video (screen recording, captions, voice), 15 min 22 s	v1.6/CTDMS_v1.6_Guide_Video.mp4
Transcript with time stamps, subtitles	v1.6/CTDMS_v1.6_Guide_Video_Transcript.txt, ..._Subtitles.srt
Demonstration master workbook and invoice files	v1.6/demo_files/
Exports produced during the demonstration	v1.6/exports/
Demonstration database + starter	v1.6/demo_database/, start_demonstration_v16.bat

4. Checklist of master records created

All of these came from the one demonstration workbook, except the checker user and the local auto added later in Master Settings (to show how later additions work).

4.1 Master upload result

Sheet	Rows	Imported	Already in CTDMS	Rejected / duplicate
ERP_Systems	4	4	0	0
Locations	5	5	0	0
Companies	4	4	0	0
Depots	4	3	0	1
Customer_Types	3	2	1	0
Customers	10	8	0	2
Transporters	5	5	0	0
Local_Auto_Rates	4	4	0	0
Freight_Rates	6	6	0	0
Invoice_Code_Mappings	4	4	0	0
ERP_Mappings	7	7	0	0

4.2 Companies and depots

	Company code	Name	Recognition code	ERP	Relationship
☑	DM-ABC	ABC Pharma Ltd (Demo)	**ABC**	SAP	Principal
☑	DM-XYZ	XYZ Healthcare Ltd (Demo)	**XYZ**	MARG	Principal
☑	DM-PQR	PQR Lifesciences Ltd (Demo)	**PQR**	CBO	Principal
☑	DM-OWN	Tulsyan Distribution (Demo)	**TDL**	INTRECT	Own

	Depot code	Depot	Locality	City	Recognition code	Companies handled
☑	DM-BSPD	Bilaspur Depot (Demo)	Vyapar Vihar	Bilaspur	**BSP**	DM-ABC, DM-PQR
☑	DM-GUMA	Guma Depot (Demo)	Guma	Raipur	**GUMA**	DM-ABC, DM-XYZ, DM-OWN
☑	DM-KVR	Kamal Vihar Depot (Demo)	Kamal Vihar	Raipur	**KVR**	DM-XYZ, DM-PQR

Raipur has two depots (Guma and Kamal Vihar), each with its own code and recognition code.

4.3 Customer types and customers

	Type key	Name	Recognition code
☑	RETAILER	Retailer	**RET**
☑	STOCKIST	Stockist	**STK**
☑	DISTRIBUTOR	Distributor	**DST**
☑	HOSPITAL	Hospital	**HSP**
☑	INSTITUTION	Institution	**INS**
☑	CHAIN_PHARMACY	Chain Pharmacy	**CHN**
☑	OTHER	Other	**OTH**
☑	SUPER_STOCKIST	Super Stockist	**SST**
☑	GOVT_SUPPLY	Government Supply	**GOV**

	Code	Customer	Type	Recognition code	Location	Source ERP code
☑	DM-C025	Sharma Medical Store (Demo)	Retailer	**C025**	Raipur City	SAP 100025
☑	DM-C031	Raipur Medical Agencies (Demo)	Stockist	**C031**	Raipur City	SAP 100031
☑	DM-C040	City Care Hospital (Demo)	Hospital	**C040**	Raipur City	SAP 100040
☑	DM-C052	Bhilai Chemists (Demo)	Retailer	**C052**	Bhilai	MARG BH052
☑	DM-C063	Durg Distributors (Demo)	Distributor	**C063**	Durg	MARG DD063
☑	DM-C071	WellCare Pharmacy Chain (Demo)	Chain Pharmacy	**C071**	Bilaspur City	SAP 100071
☑	DM-C080	Bilaspur Super Stockist (Demo)	Super Stockist	**C080**	Bilaspur City	SAP 100080
☑	DM-C090	District Govt Store Korba (Demo)	Government Supply	**C090**	Korba	SAP 100090

4.4 Locations, transporters, local autos

	Code	Location	City
☑	DM-RPR	Raipur City	Raipur
☑	DM-BHL	Bhilai	Bhilai
☑	DM-DRG	Durg	Durg
☑	DM-BSP	Bilaspur City	Bilaspur
☑	DM-KRB	Korba	Korba

	Code	Long-distance transporter	Payment
☑	DM-SRL	Shree Roadlines (Demo)	To be billed
☑	DM-VTC	Vijay Transport Co (Demo)	To be billed

	Code	Local auto	Charging	Usual trip	Rate / box	Vehicle
☑	DM-LA1	Sharma Local Auto (Demo)	Per trip	₹500.00	-	CG04-AB-1001
☑	DM-LA2	Ravi Tempo (Demo)	Per box	-	₹20.00	CG04-AB-1002
☑	DM-LA3	Kishore Mini Truck (Demo)	Per trip or per box	₹600.00	₹18.00	CG04-AB-1003
☑	DM-LA4	Mohan E-Rickshaw (Demo)	Per trip	₹300.00	-	CG04-ER-1004

	Rate code	Local auto	Location	Company	Depot	Rate / box
☑	DM-LA2-RPR	Ravi Tempo (Demo)	Raipur City	all	all	₹20.00

	Rate code	Local auto	Location	Company	Depot	Rate / box
☑	DM-LA2-BHL	Ravi Tempo (Demo)	Bhilai	all	all	₹25.00
☑	DM-LA2-PQR-RPR	Ravi Tempo (Demo)	Raipur City	DM-PQR	all	₹22.00
☑	DM-LA3-RPR	Kishore Mini Truck (Demo)	Raipur City	all	all	₹18.00

4.5 Rates, invoice markings, ERP mappings, users

	Transporter	Location	Rate / case	Status
☑	Shree Roadlines (Demo)	Bilaspur City	₹22.00	Approved
☑	Shree Roadlines (Demo)	Korba	₹30.00	Approved
☑	Shree Roadlines (Demo)	Durg	₹14.00	Approved
☑	Shree Roadlines (Demo)	Bhilai	₹15.00	Approved
☑	Vijay Transport Co (Demo)	Bilaspur City	₹20.00	Approved
☑	Vijay Transport Co (Demo)	Korba	₹28.00	Approved

	Mapping	ERP	Marking in the ERP invoice number	Company	Depot
☑	DM-MK-XYG	MARG	**XY/G/**	DM-XYZ	DM-GUMA
☑	DM-MK-XYK	MARG	**XY/K/**	DM-XYZ	DM-KVR
☑	DM-MK-PQK	CBO	**PQR/K/**	DM-PQR	DM-KVR
☑	DM-MK-PQB	CBO	**PQR/B/**	DM-PQR	DM-BSPD

ERP code mappings created: 15 (SAP plants → depots, MARG and CBO customer codes → customers). Users: admin, demo.checker.

5. All information given and used as samples

5.1 Files

File (v1.6/demo_files)	What it is	What it demonstrates
DEMO_v1.6_Master_Workbook.xlsx	The one-time master workbook (Instructions + all sheets)	Validation, preview, rejected rows (depot without city, customer typed twice, unknown location), existing code skipped
DEMO_v1.6_SAP_billing_ABC.xlsx	ABC Pharma's SAP billing export (6 invoices)	Depot from the SAP plant code, customer from SAP's customer number
DEMO_v1.6_MARG_sales_XYZ.csv	XYZ Healthcare's MARG sales register (4 invoices)	Company AND depot recognised from the marking XY/G/ or XY/K/ in the bill number
DEMO_v1.6_CBO_invoices_PQR.csv	PQR Lifesciences' CBO export (3 invoices)	Marking PQR/K/ or PQR/B/; CBO customer codes through ERP mappings

5.2 Every invoice as it stands at the end of the demonstration

Original ERP no	Invoice Recognition Code	Source · company	Depot · city	Customer · type	Location	Boxes · value	Status	Carrier	Freight (share)
001245	**ABC-RET-C025-GUMA-001245**	SAP · ABC Pharma Ltd (Demo)	Guma Depot (Demo) · Raipur	Sharma Medical Store (Demo) · Retailer	Raipur City	5 · ₹18,450.00	Gone out	Sharma Local Auto (Demo)	₹83.33
001246	**ABC-STK-C031-GUMA-001246**	SAP · ABC Pharma Ltd (Demo)	Guma Depot (Demo) · Raipur	Raipur Medical Agencies (Demo) · Stockist	Raipur City	15 · ₹52,300.00	Gone out	Sharma Local Auto (Demo)	₹250.00
001247	**ABC-HSP-C040-GUMA-001247**	SAP · ABC Pharma Ltd (Demo)	Guma Depot (Demo) · Raipur	City Care Hospital (Demo) · Hospital	Raipur City	6 · ₹21,600.00	Gone out	Sharma Local Auto (Demo)	₹100.00
001248	**ABC-CHN-C071-BSP-001248**	SAP · ABC Pharma Ltd (Demo)	Bilaspur Depot (Demo) · Bilaspur	WellCare Pharmacy Chain (Demo) · Chain Pharmacy	Bilaspur City	12 · ₹39,900.00	Gone out	Shree Roadlines (Demo)	₹264.00
001249	**ABC-SST-C080-BSP-001249**	SAP · ABC Pharma Ltd (Demo)	Bilaspur Depot (Demo) · Bilaspur	Bilaspur Super Stockist (Demo) · Super Stockist	Bilaspur City	20 · ₹68,200.00	Gone out	Shree Roadlines (Demo)	₹440.00
001250	**ABC-GOV-C090-BSP-001250**	SAP · ABC Pharma Ltd (Demo)	Bilaspur Depot (Demo) · Bilaspur	District Govt Store Korba (Demo) · Government Supply	Korba	8 · ₹27,750.00	Waiting	-	-
XY/G/2601	**XYZ-RET-C025-GUMA-002601**	MARG · XYZ Healthcare Ltd (Demo)	Guma Depot (Demo) · Raipur	Sharma Medical Store (Demo) · Retailer	Raipur City	4 · ₹14,200.00	Gone out	Sharma Local Auto (Demo)	₹66.67
XY/G/2602	**XYZ-RET-C052-GUMA-002602**	MARG · XYZ Healthcare Ltd (Demo)	Guma Depot (Demo) · Raipur	Bhilai Chemists (Demo) · Retailer	Bhilai	10 · ₹33,500.00	Gatepass printed	Kishore Mini Truck (Demo)	₹550.00
XY/K/2603	**XYZ-STK-C031-KVR-002603**	MARG · XYZ Healthcare Ltd (Demo)	Kamal Vihar Depot (Demo) · Raipur	Raipur Medical Agencies (Demo) · Stockist	Raipur City	15 · ₹47,800.00	Gone out	Ravi Tempo (Demo)	₹300.00
XY/K/2604	**XYZ-DST-C063-KVR-002604**	MARG · XYZ Healthcare Ltd (Demo)	Kamal Vihar Depot (Demo) · Raipur	Durg Distributors (Demo) · Distributor	Durg	9 · ₹29,900.00	Gone out	Shree Roadlines (Demo)	₹126.00
PQR/K/0311	**PQR-STK-C031-KVR-000311**	CBO · PQR Lifesciences Ltd (Demo)	Kamal Vihar Depot (Demo) · Raipur	Raipur Medical Agencies (Demo) · Stockist	Raipur City	6 · ₹19,800.00	Gone out	Ravi Tempo (Demo)	₹132.00
PQR/K/0312	**PQR-RET-C052-KVR-000312**	CBO · PQR Lifesciences Ltd (Demo)	Kamal Vihar Depot (Demo) · Raipur	Bhilai Chemists (Demo) · Retailer	Bhilai	7 · ₹24,100.00	Gone out	Shree Roadlines (Demo)	₹105.00
PQR/B/0313	**PQR-CHN-C071-BSP-000313**	CBO · PQR Lifesciences Ltd (Demo)	Bilaspur Depot (Demo) · Bilaspur	WellCare Pharmacy Chain (Demo) · Chain Pharmacy	Bilaspur City	10 · ₹31,200.00	Gone out	Shree Roadlines (Demo)	₹220.00

5.3 Gatepasses

Gatepass	Depot	How	Carrier	Invoices	Boxes	Owed to carrier	Booked on invoices	State
DS/DM-BSPD/2627/00001	Bilaspur Depot (Demo)	Long-distance transporter	Shree Roadlines (Demo)	3	42	₹924.00	₹924.00	Gone out
DS/DM-GUMA/2627/00001	Guma Depot (Demo)	Local auto Per trip	Sharma Local Auto (Demo)	4	30	₹500.00	₹500.00	Gone out
DS/DM-KVR/2627/00001	Kamal Vihar Depot (Demo)	Local auto Per box	Ravi Tempo (Demo)	2	21	₹432.00	₹432.00	Gone out
DS/DM-GUMA/2627/00002	Guma Depot (Demo)	Local auto Per trip	Kishore Mini Truck (Demo)	1	10	₹550.00	₹550.00	Printed
DS/DM-KVR/2627/00002	Kamal Vihar Depot (Demo)	Long-distance transporter	Shree Roadlines (Demo)	2	16	₹231.00	₹231.00	Gone out

For every local-auto trip the two money columns are equal: the trip amount is shared over the invoices for the MIS and never multiplied.

6. Every invoice and trip explained in plain language

001245 ABC Pharma Ltd (Demo) → Sharma Medical Store (Demo) (Retailer), Raipur City · 5 boxes · ₹18,450.00

What it is: SAP invoice billed from Guma Depot (Demo) (Raipur). Company chosen for the SAP file; depot from the SAP plant code.
Code: Recognition code **ABC-RET-C025-GUMA-001245** = ABC company · Retailer · customer DM-C025 · Guma Depot (Demo).
Dispatch and cost: Carried by local auto Sharma Local Auto (Demo) (per trip ₹500 for the whole trip; this invoice's share ₹83.33).
Why it matters: The example management asked for: original number **001245**, Invoice Recognition Code **ABC-RET-C025-GUMA-001245**. The number was not changed in any way.

001246 ABC Pharma Ltd (Demo) → Raipur Medical Agencies (Demo) (Stockist), Raipur City · 15 boxes · ₹52,300.00

What it is: SAP invoice billed from Guma Depot (Demo) (Raipur). Company chosen for the SAP file; depot from the SAP plant code.
Code: Recognition code **ABC-STK-C031-GUMA-001246** = ABC company · Stockist · customer DM-C031 · Guma Depot (Demo).
Dispatch and cost: Carried by local auto Sharma Local Auto (Demo) (per trip ₹500 for the whole trip; this invoice's share ₹250.00).

001247 ABC Pharma Ltd (Demo) → City Care Hospital (Demo) (Hospital), Raipur City · 6 boxes · ₹21,600.00

What it is: SAP invoice billed from Guma Depot (Demo) (Raipur). Company chosen for the SAP file; depot from the SAP plant code.
Code: Recognition code **ABC-HSP-C040-GUMA-001247** = ABC company · Hospital · customer DM-C040 · Guma Depot (Demo).
Dispatch and cost: Carried by local auto Sharma Local Auto (Demo) (per trip ₹500 for the whole trip; this invoice's share ₹100.00).

001248 ABC Pharma Ltd (Demo) → WellCare Pharmacy Chain (Demo) (Chain Pharmacy), Bilaspur City · 12 boxes · ₹39,900.00

What it is: SAP invoice billed from Bilaspur Depot (Demo) (Bilaspur). Company chosen for the SAP file; depot from the SAP plant code.
Code: Recognition code **ABC-CHN-C071-BSP-001248** = ABC company · Chain Pharmacy · customer DM-C071 · Bilaspur Depot (Demo).
Dispatch and cost: Sent with long-distance transporter Shree Roadlines (Demo) from the rate card (₹264.00), bill later.

001249 ABC Pharma Ltd (Demo) → Bilaspur Super Stockist (Demo) (Super Stockist), Bilaspur City · 20 boxes · ₹68,200.00

What it is: SAP invoice billed from Bilaspur Depot (Demo) (Bilaspur). Company chosen for the SAP file; depot from the SAP plant code.
Code: Recognition code **ABC-SST-C080-BSP-001249** = ABC company · Super Stockist · customer DM-C080 · Bilaspur Depot (Demo).
Dispatch and cost: Sent with long-distance transporter Shree Roadlines (Demo) from the rate card (₹440.00), bill later.

001250 ABC Pharma Ltd (Demo) → District Govt Store Korba (Demo) (Government Supply), Korba · 8 boxes · ₹27,750.00

What it is: SAP invoice billed from Bilaspur Depot (Demo) (Bilaspur). Company chosen for the SAP file; depot from the SAP plant code.
Code: Recognition code **ABC-GOV-C090-BSP-001250** = ABC company · Government Supply · customer DM-C090 · Bilaspur Depot (Demo).
Dispatch and cost: Not dispatched yet.
Why it matters: Its customer type, Government Supply, had no short code, so at first no code was made and the reason was shown. After the code GOV was entered in Master Settings the code appeared at once. It is the invoice still waiting for dispatch.

XY/G/2601 XYZ Healthcare Ltd (Demo) → Sharma Medical Store (Demo) (Retailer), Raipur City · 4 boxes · ₹14,200.00

What it is: MARG invoice billed from Guma Depot (Demo) (Raipur). Company and depot recognised from the marking **XY/G/** in the ERP number.
Code: Recognition code **XYZ-RET-C025-GUMA-002601** = XYZ company · Retailer · customer DM-C025 · Guma Depot (Demo).
Dispatch and cost: Carried by local auto Sharma Local Auto (Demo) (per trip ₹500 for the whole trip; this invoice's share ₹66.67).

XY/G/2602 XYZ Healthcare Ltd (Demo) → Bhilai Chemists (Demo) (Retailer), Bhilai · 10 boxes · ₹33,500.00

What it is: MARG invoice billed from Guma Depot (Demo) (Raipur). Company and depot recognised from the marking **XY/G/** in the ERP number.
Code: Recognition code **XYZ-RET-C052-GUMA-002602** = XYZ company · Retailer · customer DM-C052 · Guma Depot (Demo).
Dispatch and cost: Carried by local auto Kishore Mini Truck (Demo) (per trip ₹550 for the whole trip; this invoice's share ₹550.00).
Why it matters: Kishore Mini Truck can be charged either way; today per trip, ₹550 for this one invoice.

XY/K/2603 XYZ Healthcare Ltd (Demo) → Raipur Medical Agencies (Demo) (Stockist), Raipur City · 15 boxes · ₹47,800.00

What it is: MARG invoice billed from Kamal Vihar Depot (Demo) (Raipur). Company and depot recognised from the marking **XY/K/** in the ERP number.
Code: Recognition code **XYZ-STK-C031-KVR-002603** = XYZ company · Stockist · customer DM-C031 · Kamal Vihar Depot (Demo).
Dispatch and cost: Carried by local auto Ravi Tempo (Demo) (per box; ₹300.00).
Why it matters: 15 boxes at Ravi Tempo's Raipur rate of ₹20 = ₹300 (the per-box example).

XY/K/2604 XYZ Healthcare Ltd (Demo) → Durg Distributors (Demo) (Distributor), Durg · 9 boxes · ₹29,900.00

What it is: MARG invoice billed from Kamal Vihar Depot (Demo) (Raipur). Company and depot recognised from the marking **XY/K/** in the ERP number.
Code: Recognition code **XYZ-DST-C063-KVR-002604** = XYZ company · Distributor · customer DM-C063 · Kamal Vihar Depot (Demo).
Dispatch and cost: Sent with long-distance transporter Shree Roadlines (Demo) from the rate card (₹126.00), bill later.
Why it matters: Was added to Ravi Tempo's per-box trip, but Ravi Tempo has no per-box rate for Durg, so CTDMS refused it. It then went with Shree Roadlines in the keyboard-only dispatch (Step 9 of the video), together with PQR/K/0312.

PQR/K/0311 PQR Lifesciences Ltd (Demo) → Raipur Medical Agencies (Demo) (Stockist), Raipur City · 6 boxes · ₹19,800.00

What it is: CBO invoice billed from Kamal Vihar Depot (Demo) (Raipur). Company and depot recognised from the marking **PQR/K/** in the ERP number.
Code: Recognition code **PQR-STK-C031-KVR-000311** = PQR company · Stockist · customer DM-C031 · Kamal Vihar Depot (Demo).
Dispatch and cost: Carried by local auto Ravi Tempo (Demo) (per box; ₹132.00).
Why it matters: 6 boxes at ₹22 - Ravi Tempo's own rate for PQR's goods in Raipur, which wins over the general Raipur rate.

PQR/K/0312 PQR Lifesciences Ltd (Demo) → Bhilai Chemists (Demo) (Retailer), Bhilai · 7 boxes · ₹24,100.00

What it is: CBO invoice billed from Kamal Vihar Depot (Demo) (Raipur). Company and depot recognised from the marking **PQR/K/** in the ERP number.
Code: Recognition code **PQR-RET-C052-KVR-000312** = PQR company · Retailer · customer DM-C052 · Kamal Vihar Depot (Demo).
Dispatch and cost: Sent with long-distance transporter Shree Roadlines (Demo) from the rate card (₹105.00), bill later.
Why it matters: Dispatched in the keyboard-only demonstration: picked from the waiting list with the Down arrow and Enter, transporter found by typing its code SRL, gatepass generated, printed and completed with Enter, F8 and F9.

PQR/B/0313 PQR Lifesciences Ltd (Demo) → WellCare Pharmacy Chain (Demo) (Chain Pharmacy), Bilaspur City · 10 boxes · ₹31,200.00

What it is: CBO invoice billed from Bilaspur Depot (Demo) (Bilaspur). Company and depot recognised from the marking **PQR/B/** in the ERP number.
Code: Recognition code **PQR-CHN-C071-BSP-000313** = PQR company · Chain Pharmacy · customer DM-C071 · Bilaspur Depot (Demo).
Dispatch and cost: Sent with long-distance transporter Shree Roadlines (Demo) from the rate card (₹220.00), bill later.

The local-auto trips

DS/DM-GUMA/2627/00001 - Sharma Local Auto (Demo) carried 4 invoice(s), 30 boxes, for one trip charge of **₹500.00**. The auto is owed exactly that. For the MIS it is shared by boxes; the shares add up to ₹500.00. Charging per invoice would have shown ₹2,000 - that double counting cannot happen.

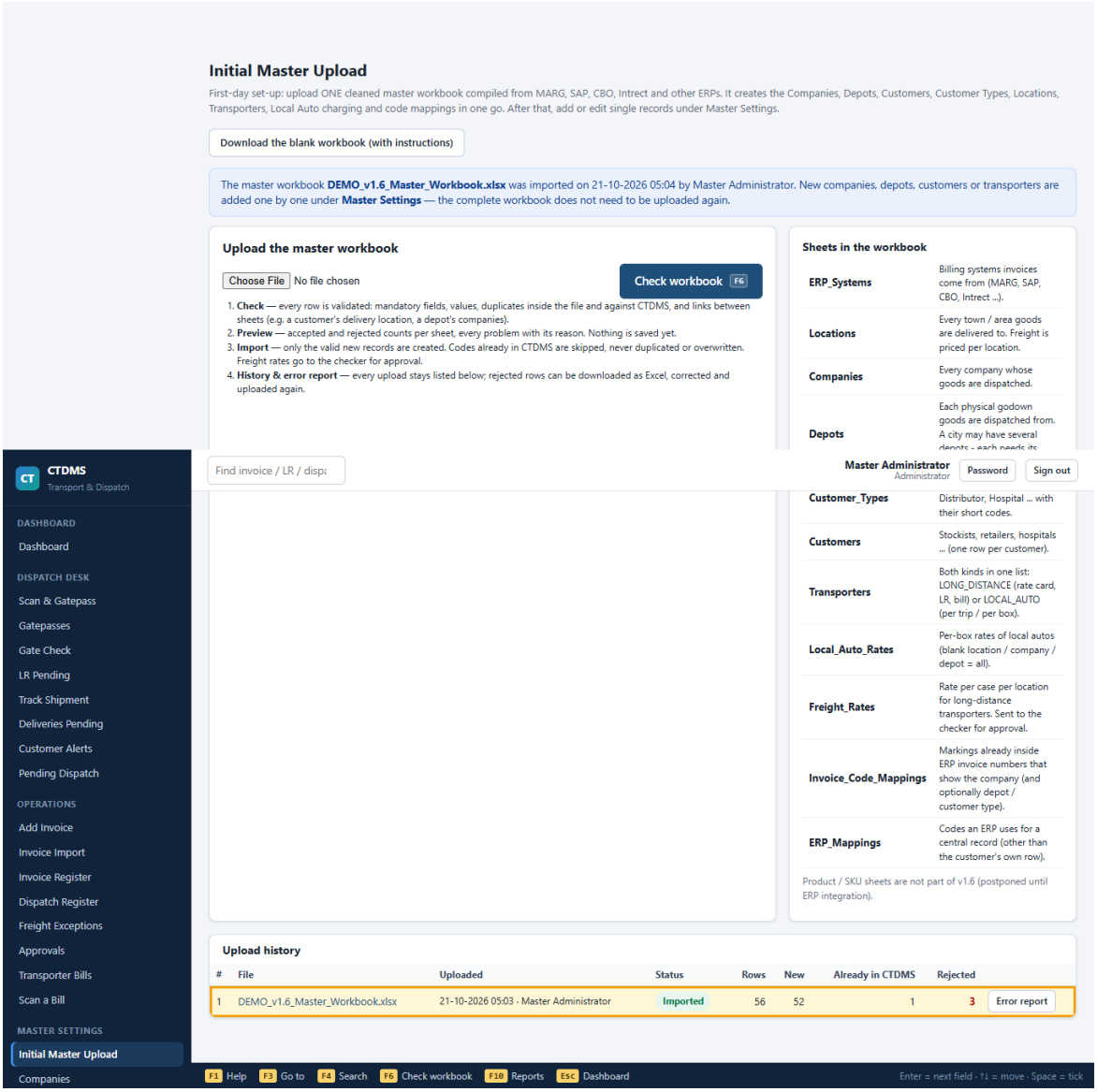
DS/DM-KVR/2627/00001 - Ravi Tempo (Demo) charged per box: 21 boxes, each invoice at its own rate, total **₹432.00**.

DS/DM-GUMA/2627/00002 - Kishore Mini Truck (Demo) carried 1 invoice(s), 10 boxes, for one trip charge of **₹550.00**. The auto is owed exactly that. For the MIS it is shared by boxes; the shares add up to ₹550.00.

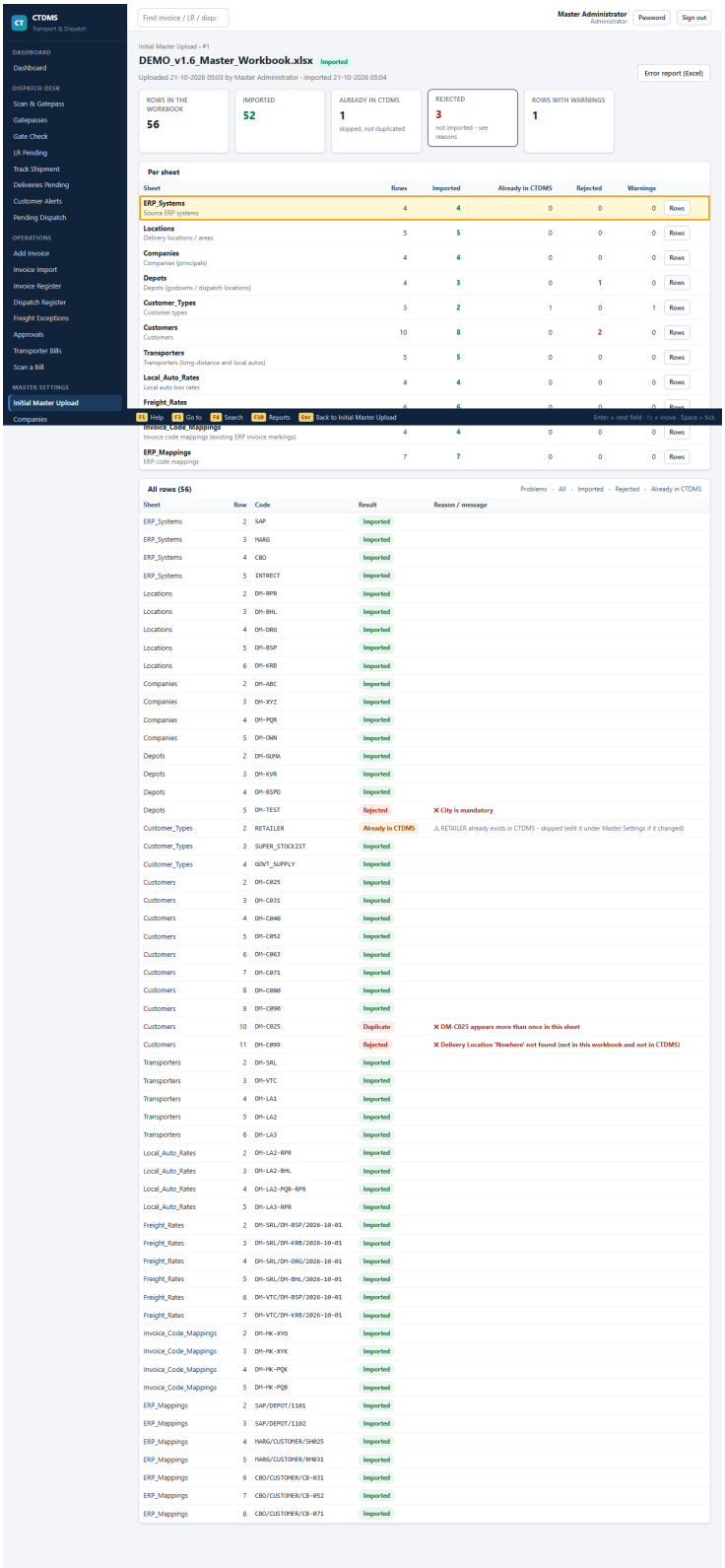
7. Step-by-step with the screens

Step 1. Initial Master Upload

1. Master Settings › Initial Master Upload › **Download the blank workbook**. Read the Instructions sheet.
2. Fill the sheets with the cleaned data from all ERPs (codes unique, mandatory columns marked *).
3. Upload › **Check workbook**. Read the per-sheet counts and every problem row.
4. **Import N valid records**. Download the error report, correct the rejected rows, upload them again if needed.
5. Ask the checker to approve the freight rates (Approvals › Approve all).

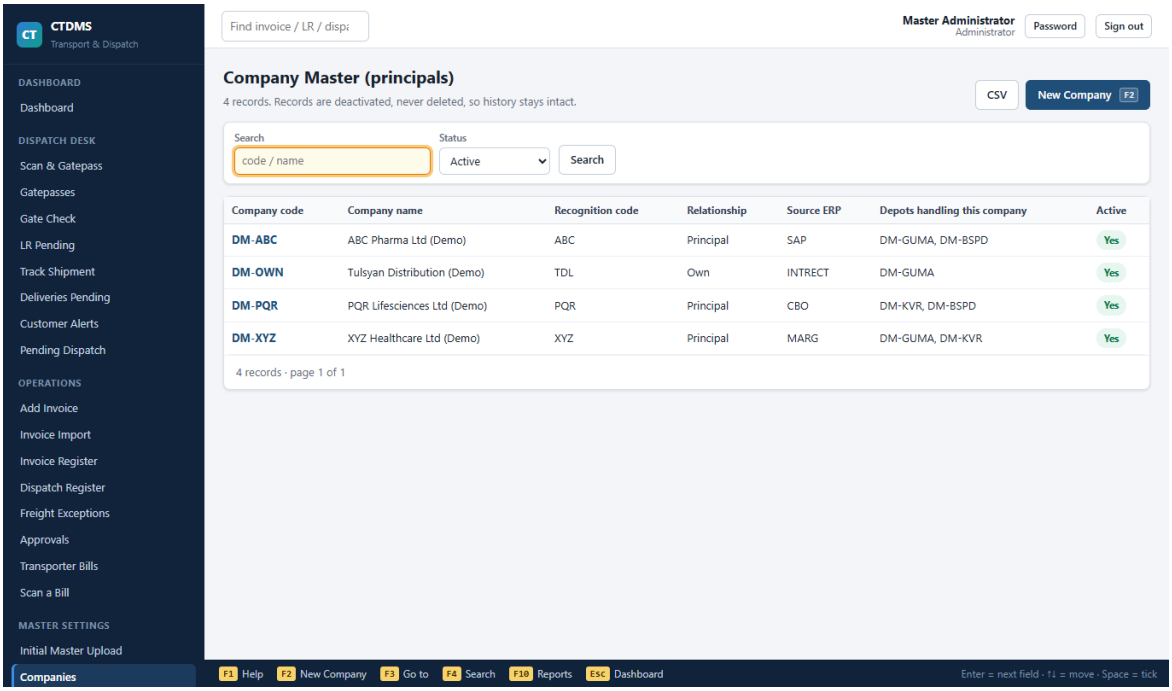


Initial Master Upload: template, steps and history.

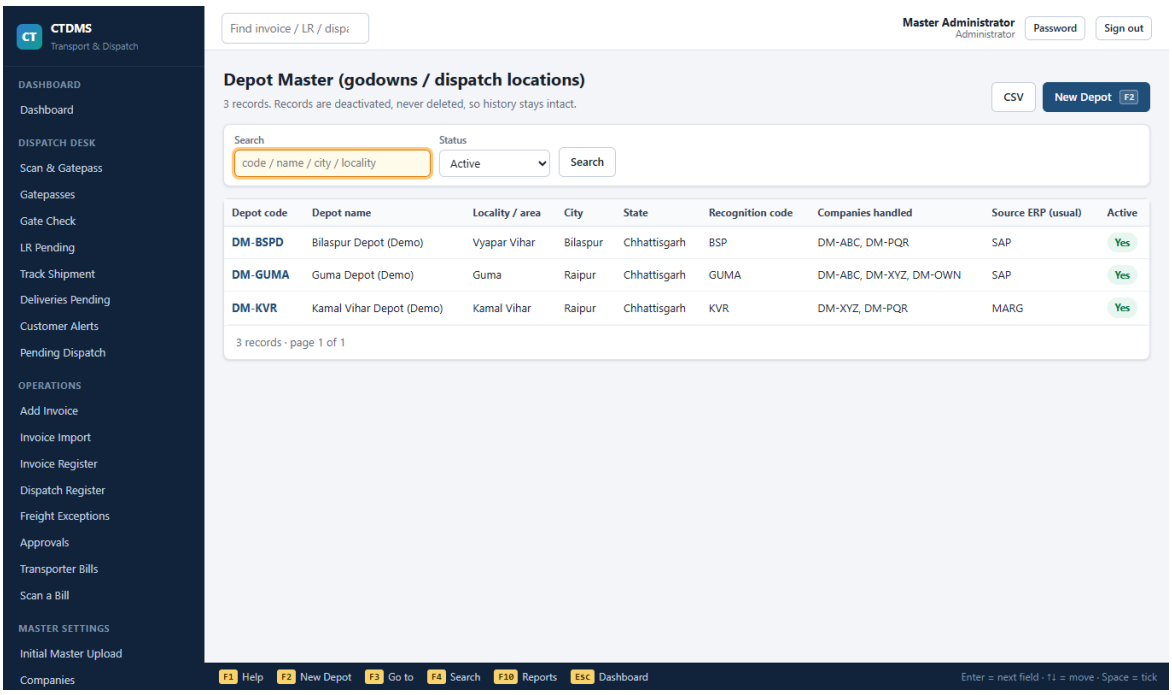


Result of the demonstration workbook: per-sheet counts and every row with its result.

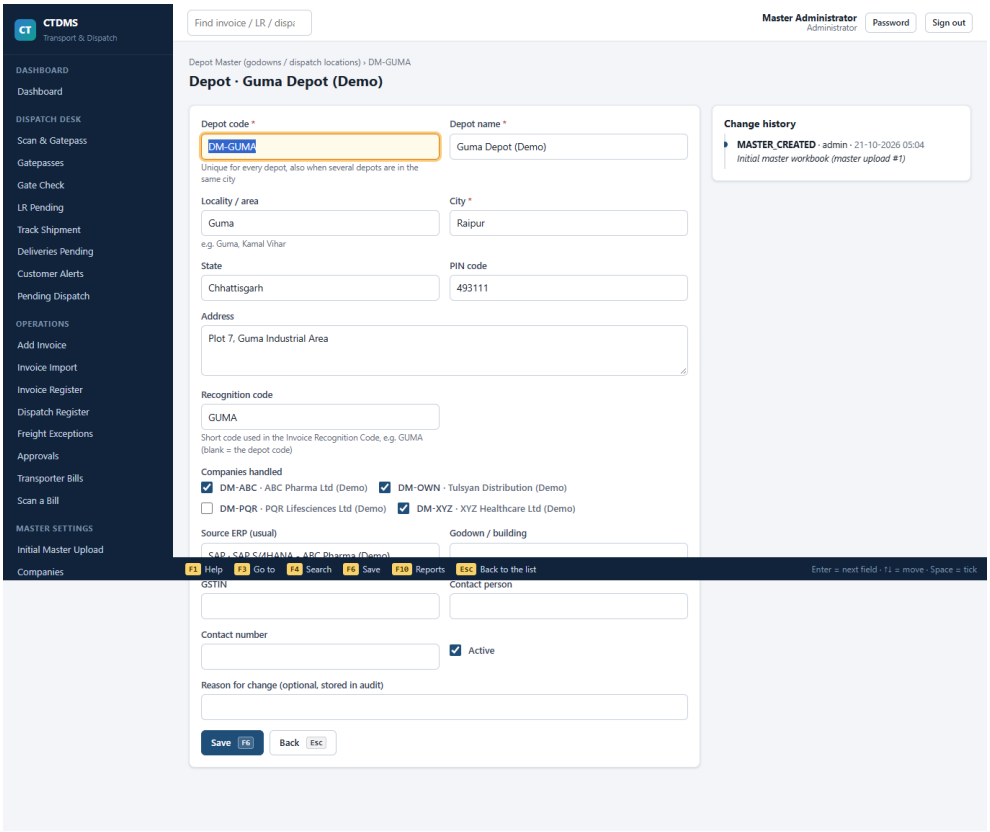
Step 2. The masters



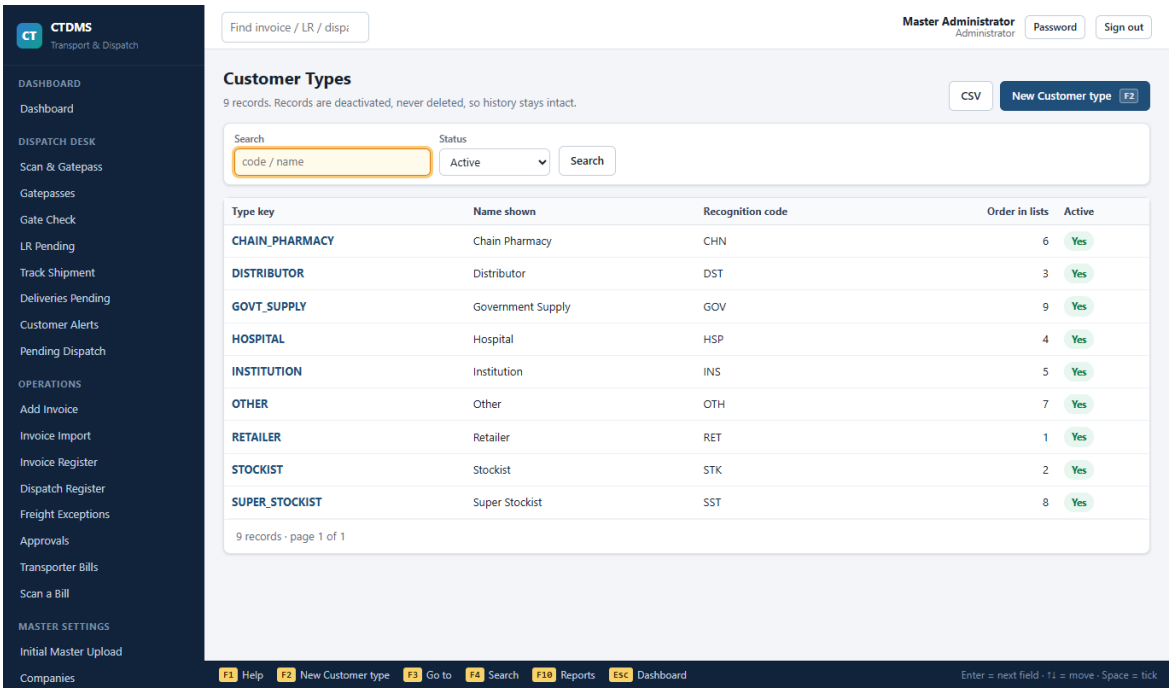
Company Master with recognition codes.



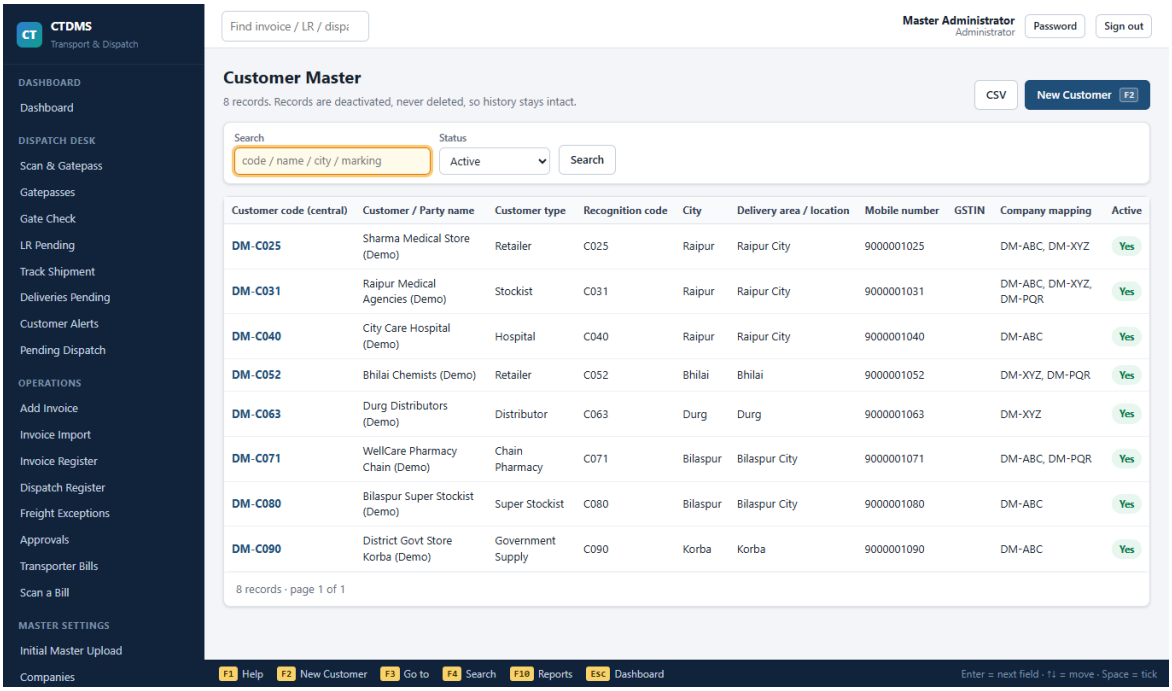
Depot Master: Guma and Kamal Vihar in Raipur, Bilaspur.



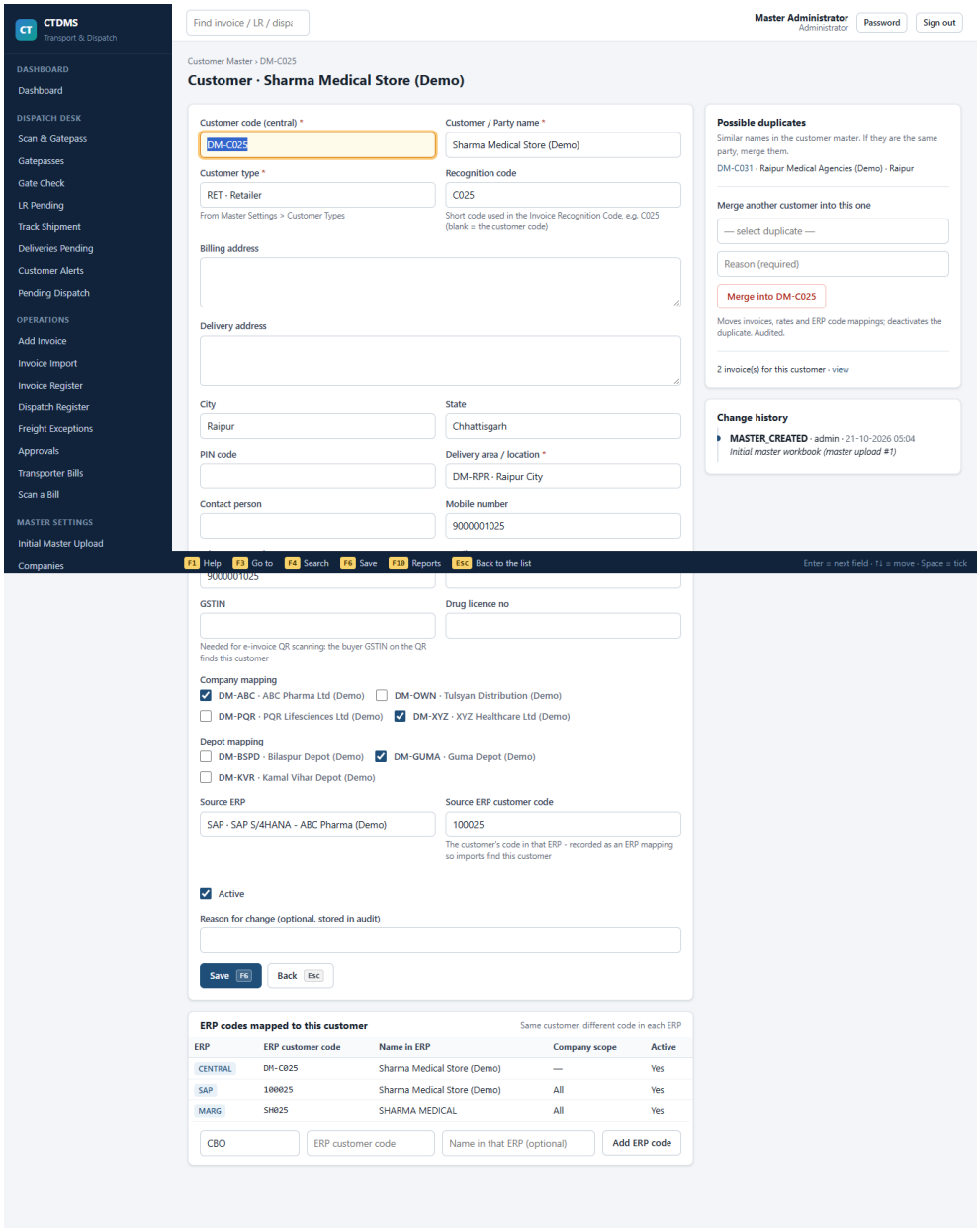
A depot: locality, city, companies handled, recognition code.



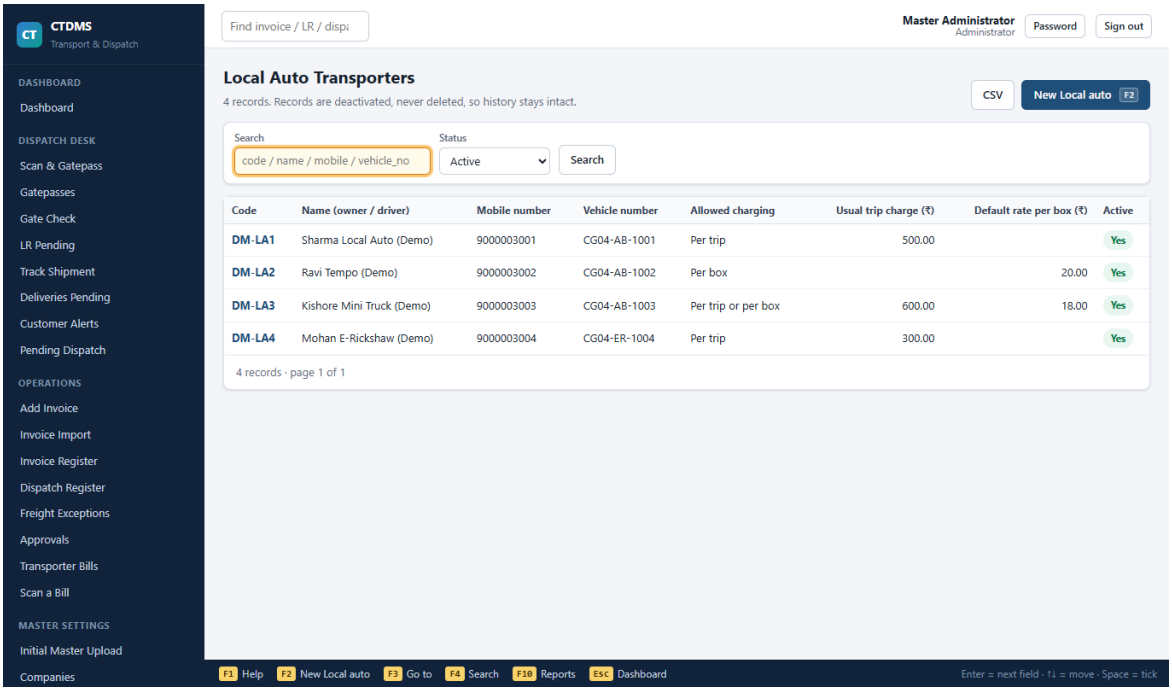
Customer Types master with short codes.



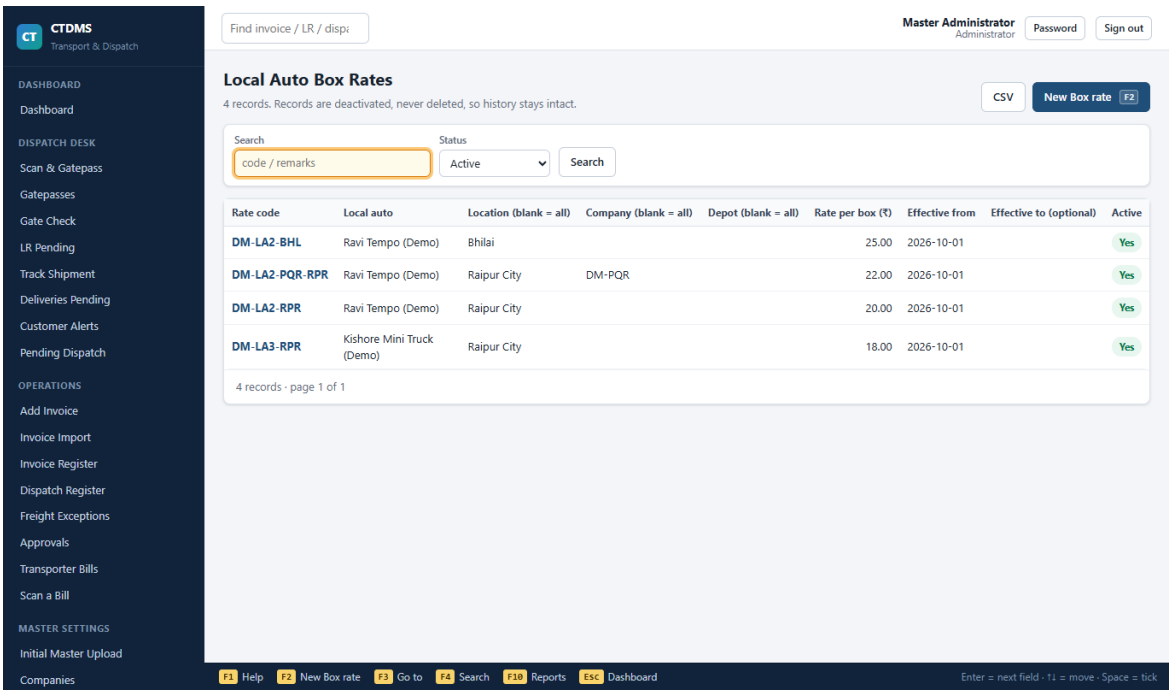
Customer Master.



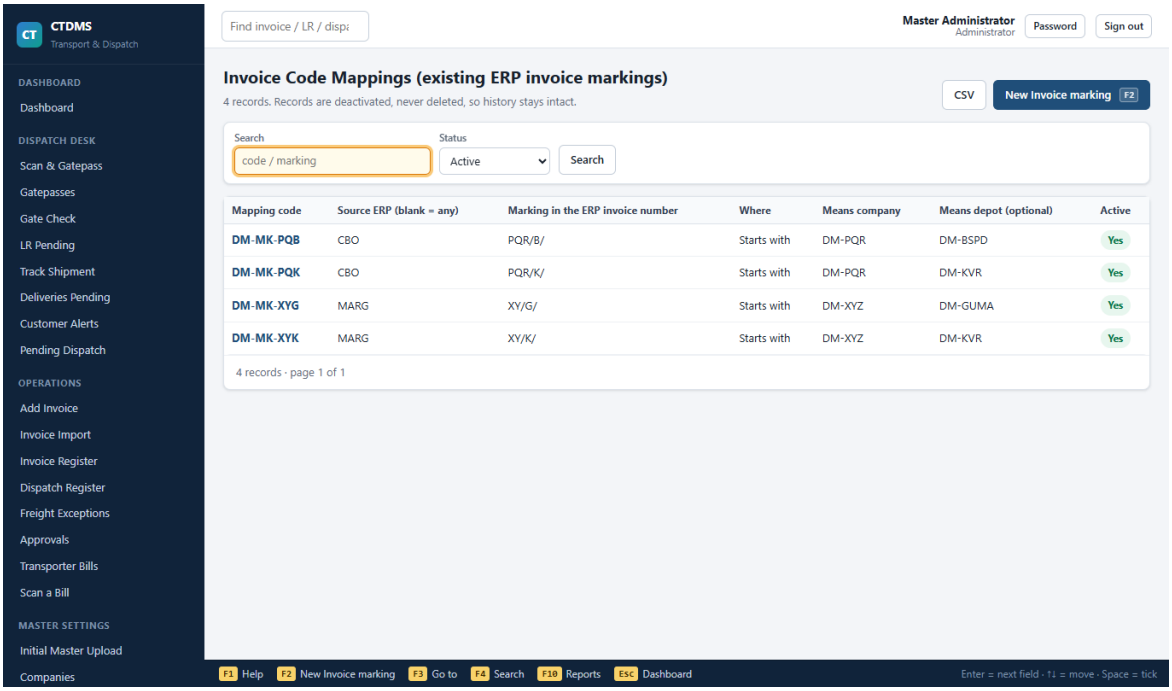
A customer: type, recognition code, source ERP and its code.



Local Auto Transporters: per trip, per box, both.



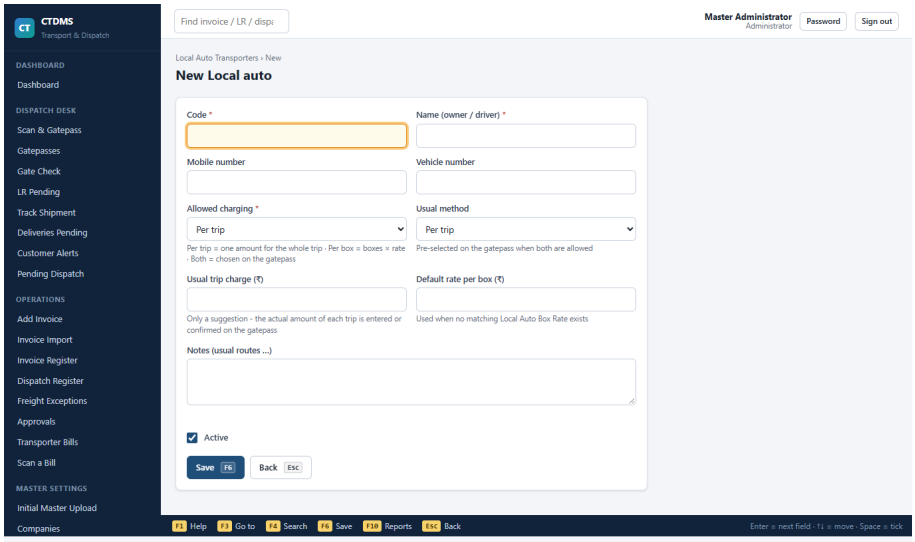
Local Auto Box Rates by location / company / depot.



Invoice Code Mappings: existing ERP invoice markings.

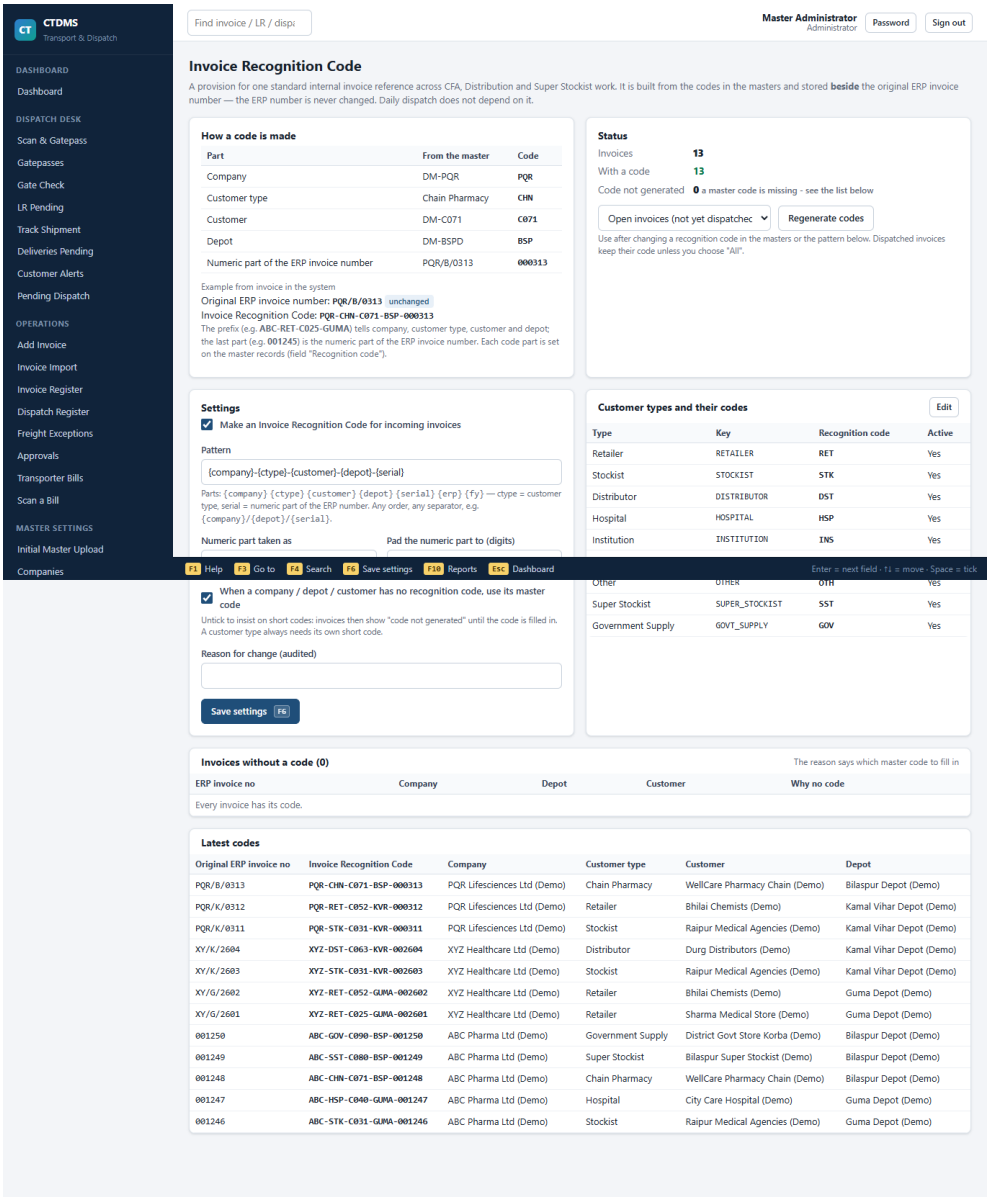
Step 3. Adding a record later

Master Settings › the master › **New**. Example: a new local auto. It appears in the gatepass dropdown at once. Records already used are made inactive, never deleted.



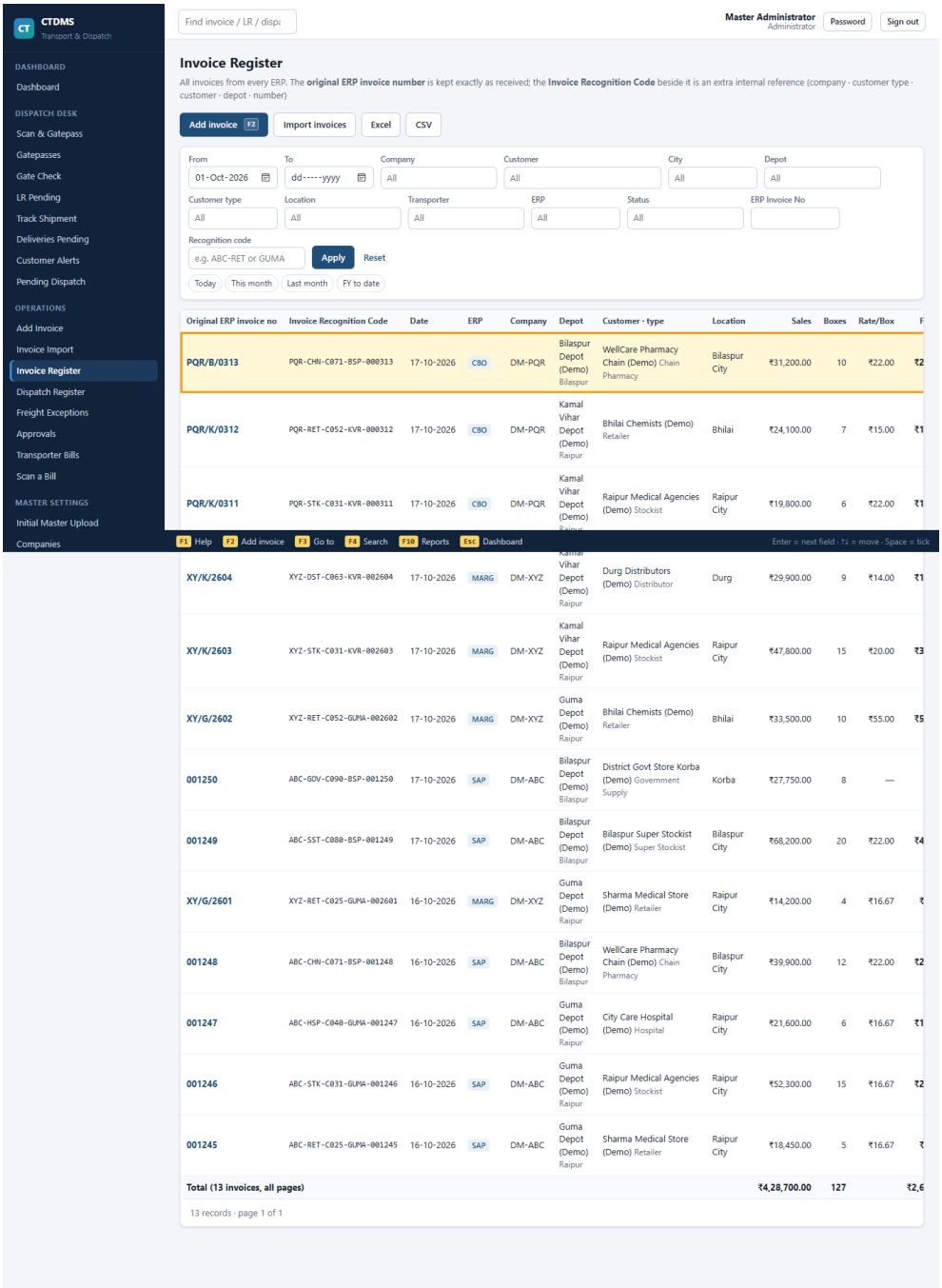
Adding a local auto.

Step 4. Invoice Recognition Code

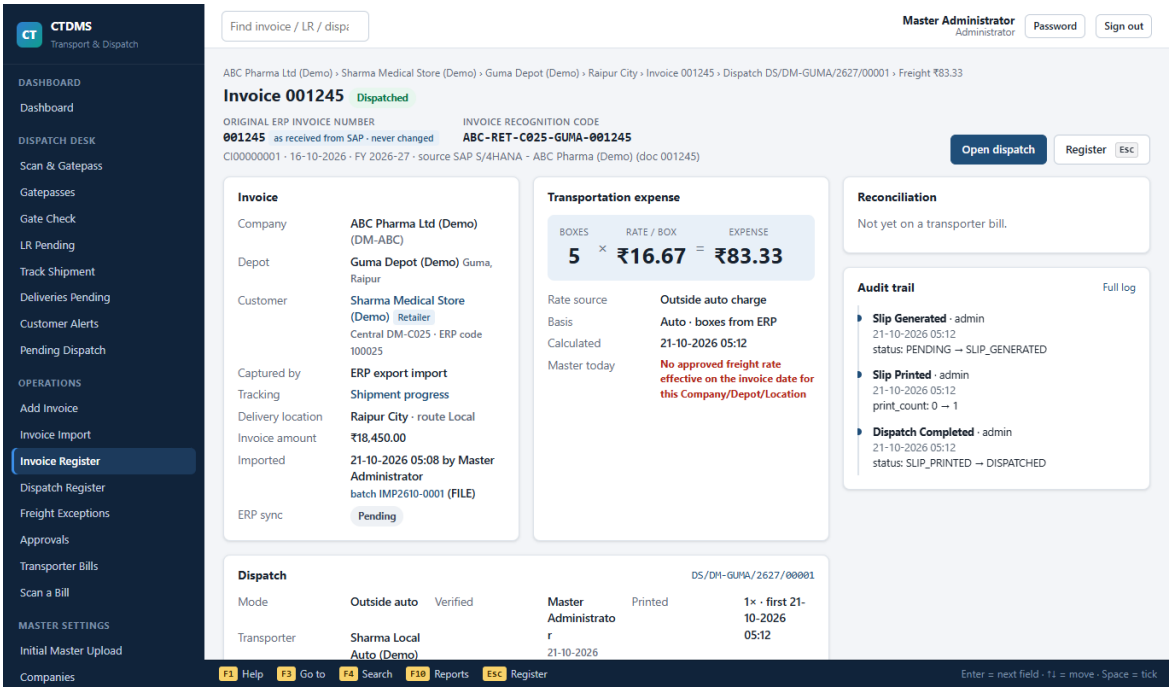


Settings, example, missing codes, regenerate.

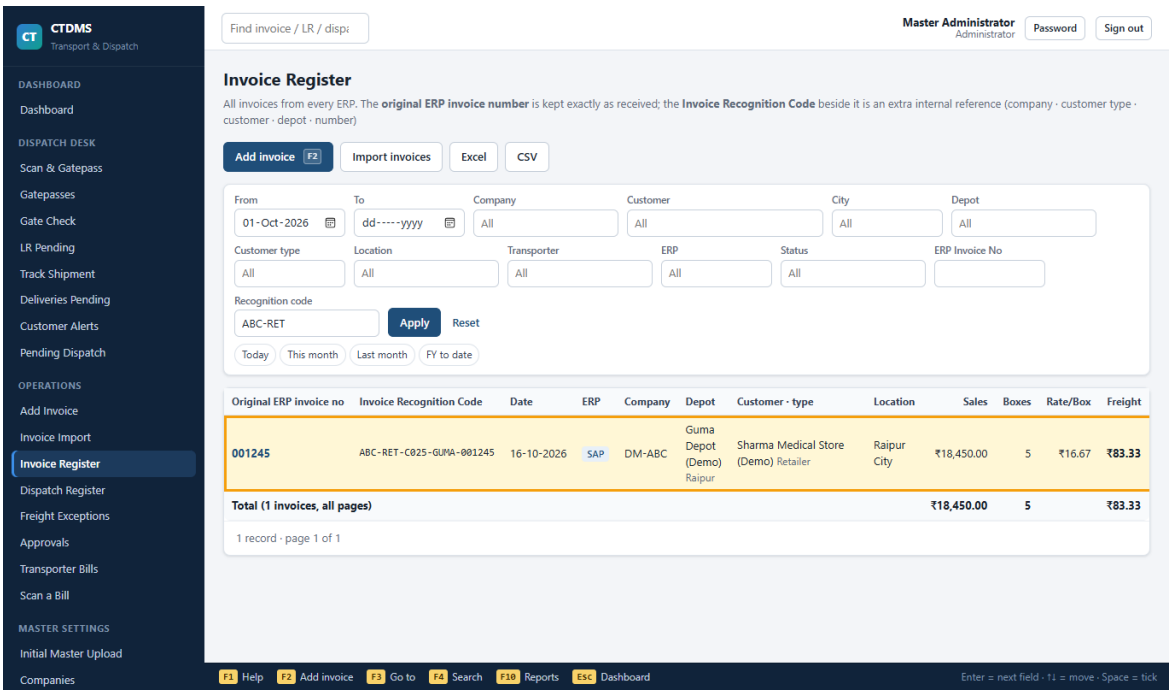
Step 5. Invoices and search



Invoice Register: original ERP number and Invoice Recognition Code side by side.

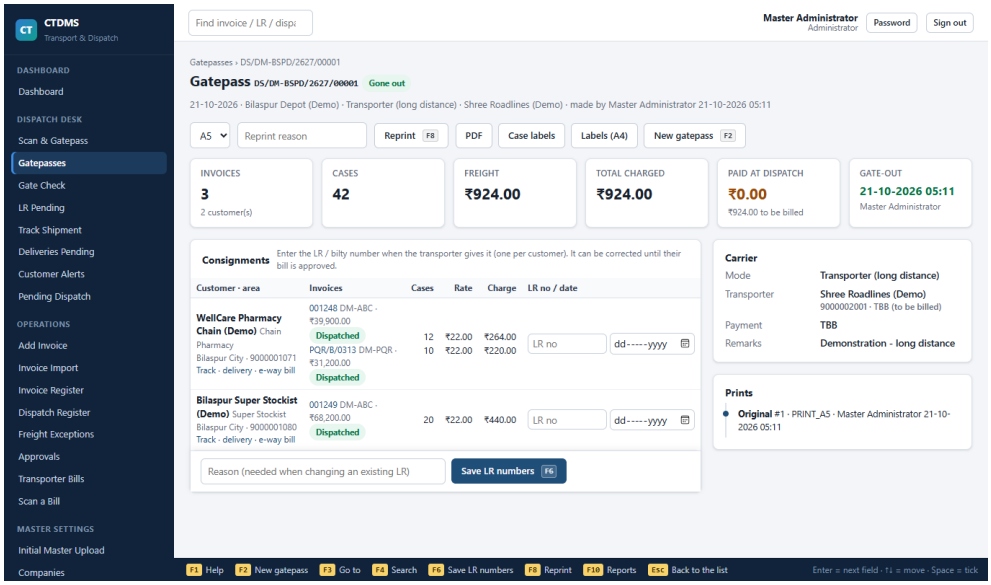


Invoice 001245 - ABC-RET-C025-GUMA-001245; the ERP number is unchanged.

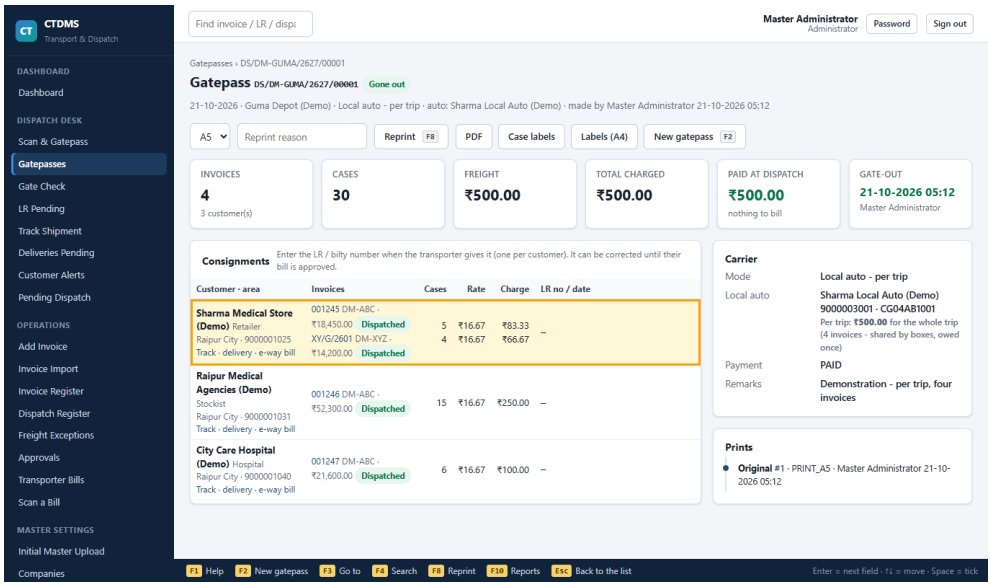


Search by the start of the code: ABC-RET = ABC Pharma invoices to retailers.

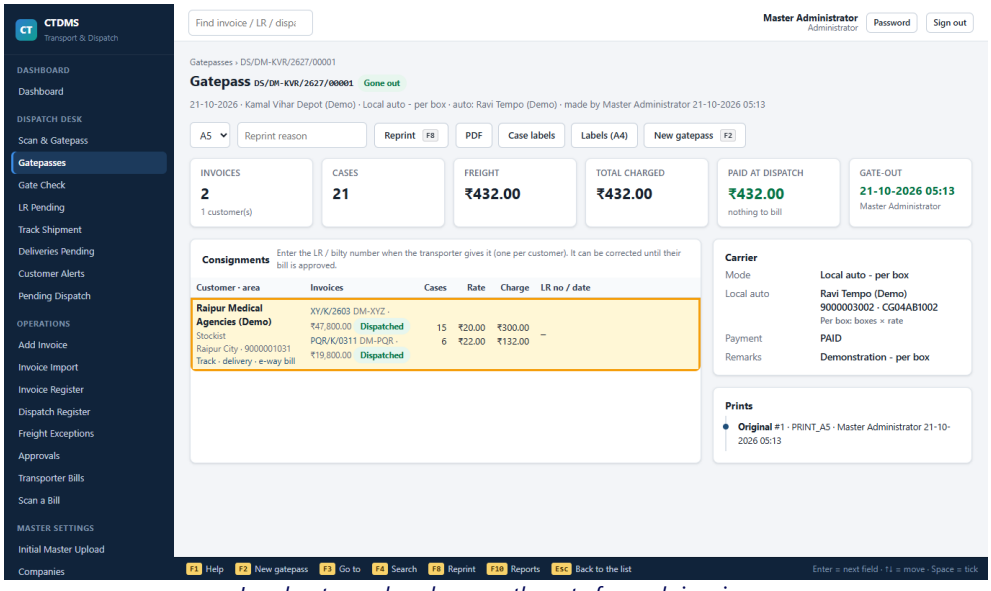
Step 6. Dispatch



Long-distance transporter: unchanged v1.5 workflow.



Local auto per trip: four invoices, one trip charge of ₹500.



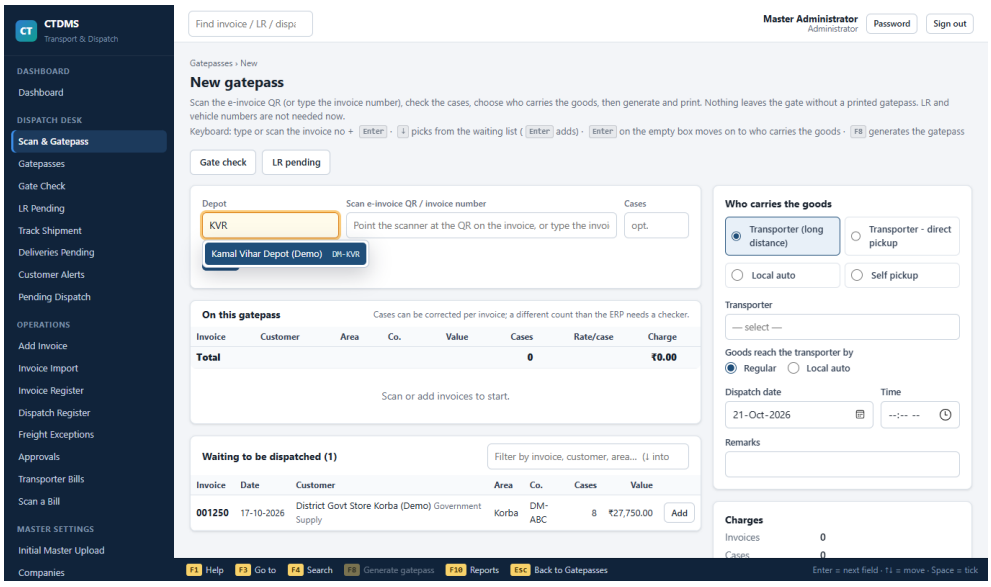
Local auto per box: boxes × the rate for each invoice.

Step 7. Keyboard-only dispatch (MARG style)

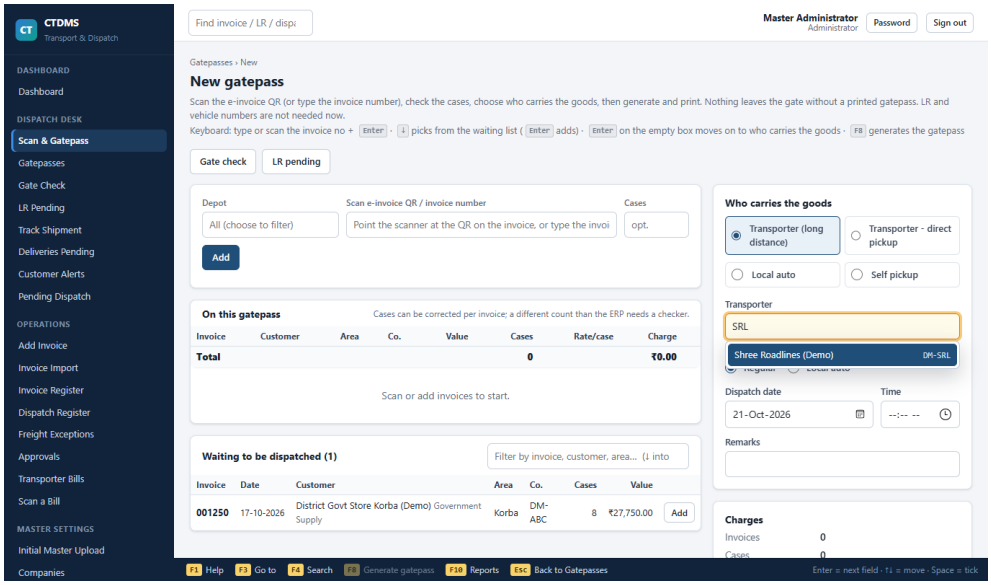
Every daily screen can be worked without the mouse. A key always means the same thing; the dark bar at the bottom of the screen shows the keys that work on that screen, and **F1** lists all of them. F5 (refresh), F11 (full screen) and F12 are left to the browser.

Key	Meaning (same on every screen)	Key	Moving around
F1	Help - list of keys	Enter	Next field; after the last field the cursor goes to the main button, Enter again presses it
F2	New / Add (dispatch desk: new gatepass)	Shift + Enter	Previous field
F3	Go to any screen by typing its name (also Ctrl+K)	Type	In a choice box: the code (GUMA, C025, SRL) or part of the name; ↓ ↑ to pick, Enter to select
F4	Search box of the screen	↓ ↑	Move through a table; ↓ from a search box goes into its table
F6	Save (also Ctrl+S)	Enter on a row	Open it (gatepass screen: add the invoice)
F7	Edit	Space	Tick / untick
F8	Generate the gatepass / dispatch slip, or print it	Esc	Close a list or question; otherwise back one screen - to the same list and row
F9	Complete the dispatch (gate-out)	/	Find an invoice, LR or gatepass from any screen
F10	Reports		

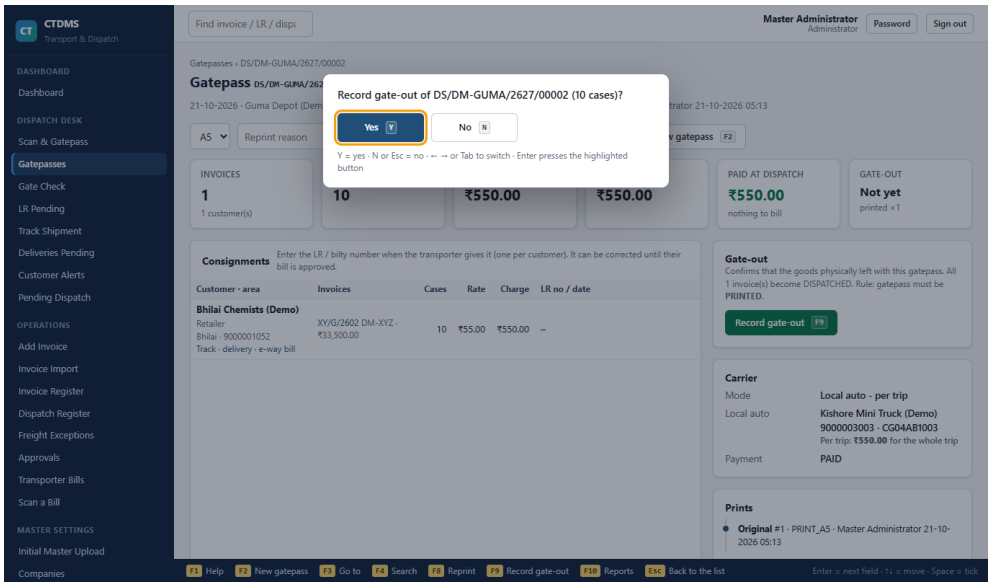
1. **F2** - New gatepass. The cursor is in Depot: type **KVR**, Enter.
2. Type or scan the invoice number, Enter. Or ↓ into the waiting list and Enter on the invoice.
3. Enter on the empty scan box - Mode of dispatch (arrows change it), Enter - Transporter: type **SRL**, Enter. For a local auto: type its code, Enter; the trip charge and vehicle come from the master, Enter, Enter.
4. Enter through date, time and remarks; the cursor lands on **Generate gatepass**; Enter.
5. **F8** prints. **F9** completes the dispatch - CTDMS asks first, Enter = Yes.
6. **Esc** back to the list; ↓ for the next record. Delete, void, cancel and reject always ask with **No** pre-selected.



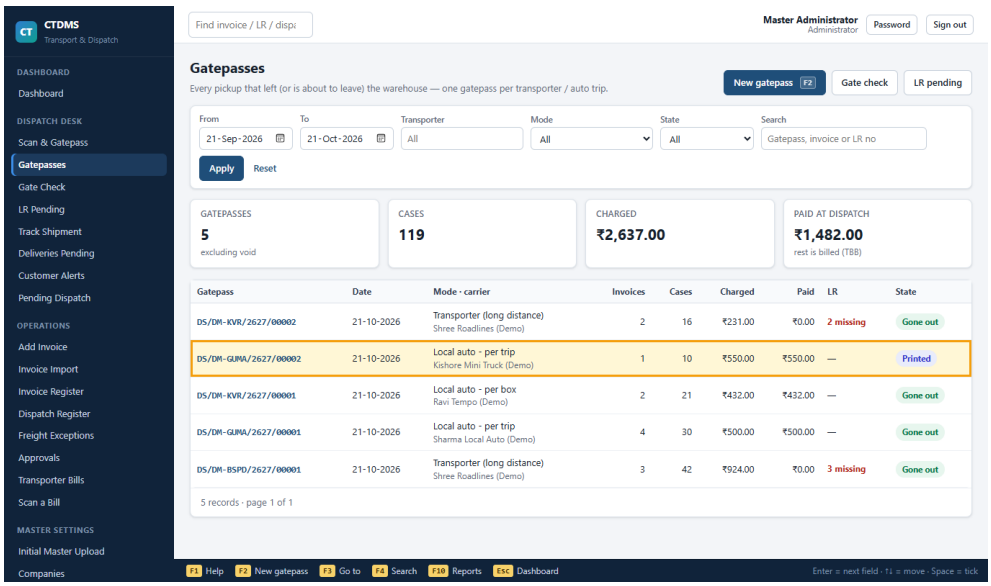
F2 - New gatepass: typing the depot code KVR finds Kamal Vihar Depot.



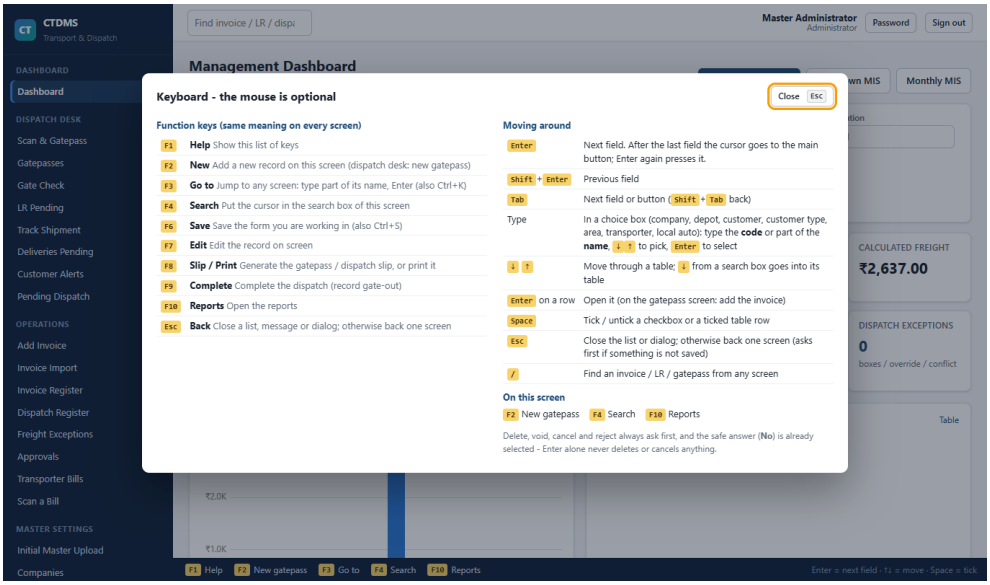
Transporter box: typing SRL finds Shree Roadlines; Enter selects and moves on.



F9 - complete the dispatch: a Yes / No question; for void / cancel the safe answer No is pre-selected.



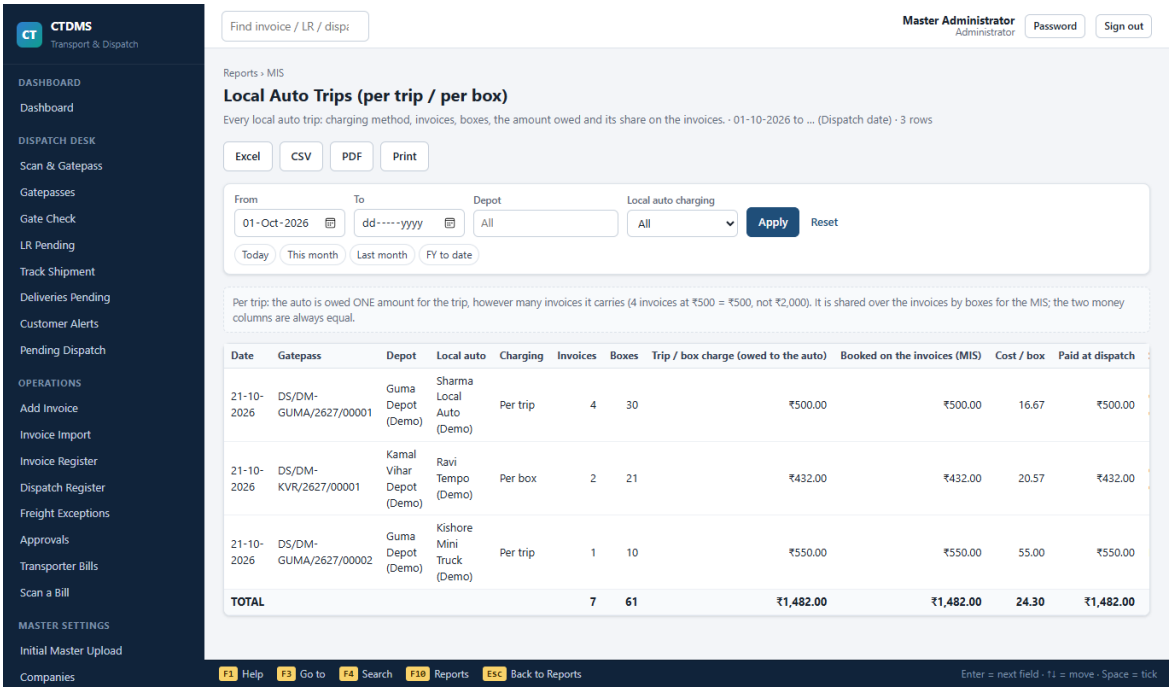
Lists: ↓ ↑ move the yellow row, Enter opens it, Esc comes back to the same row.



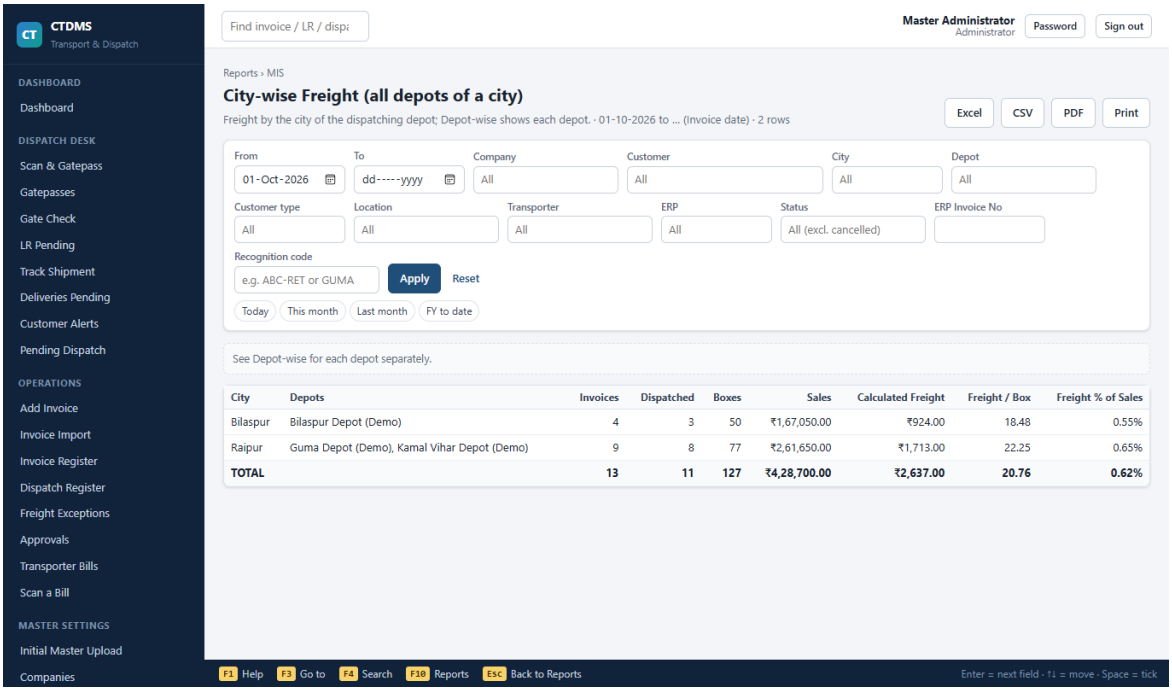
F1 - all keys, and the keys of the current screen.

TIP Tested: the whole dispatch - depot, invoices, transporter or local auto, generate, print, gate-out, back to the list - was done with the keyboard only (58 automatic checks, tests/e2e_keyboard_dispatch.py).

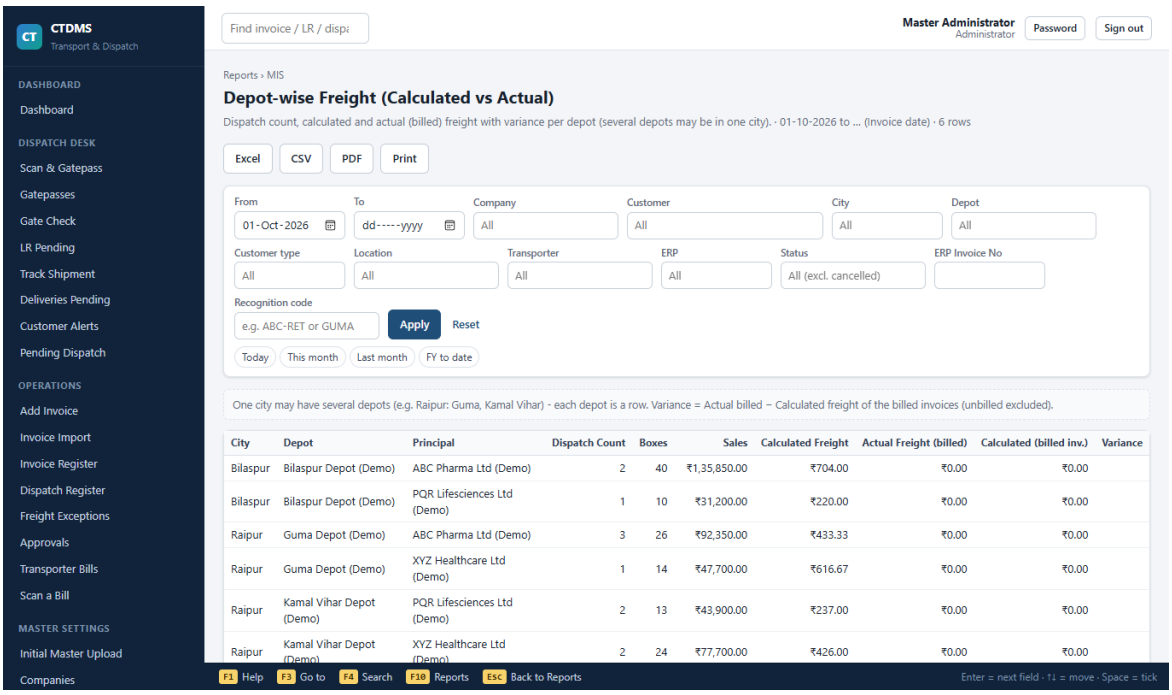
Step 8. Reports



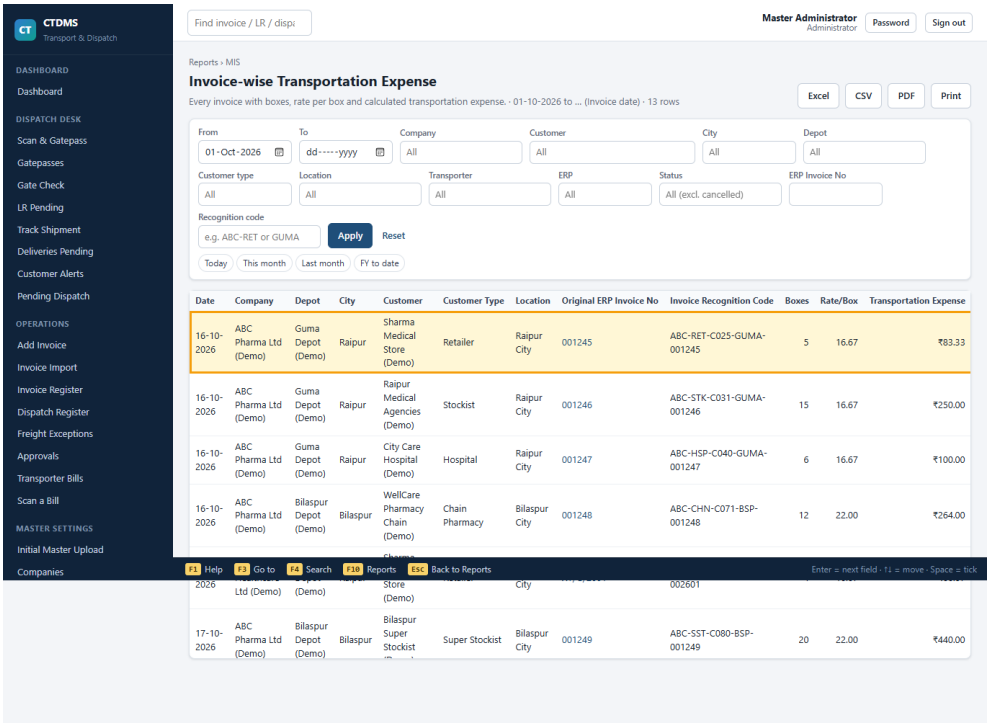
Local Auto Trips: owed to the auto = booked on the invoices.



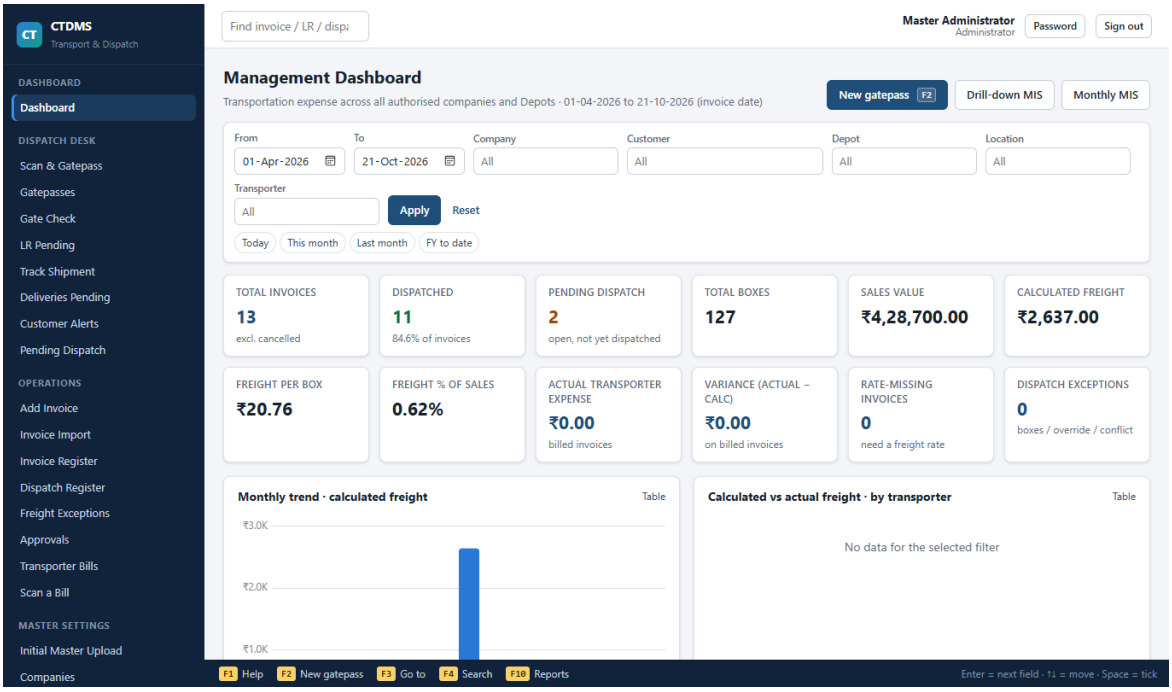
City-wise freight: all Raipur depots together.



Depot-wise freight: each depot with its city.



Invoice-wise: original ERP number and recognition code in separate columns.



Dashboard.

8. The master workbook - every column

ERP_Systems - Source ERP systems

Column	Mandatory	What it means	Allowed values	Example
ERP Code	Yes	Short unique code		MARG
ERP Name	Yes	Name shown on screens		MARG ERP 9+ (Company B)
How data arrives		FILE = Excel/CSV upload (default) · FOLDER = drop folder · API = pushed by the ERP	FILE, FOLDER, API, MARG, SAP, CBO, DATABASE	FILE
Active		Yes / No (default Yes)	Yes / No	Yes
Notes		Status of the link, contact person ...		Exported every evening

Locations - Delivery locations / areas

Column	Mandatory	What it means	Allowed values	Example
Location Code	Yes			RPR-CITY
Location Name	Yes			Raipur City
City				Raipur
State				Chhattisgarh
PIN Code				492001
Route				Local
Zone				Central
Distance km			number	0
Active			Yes / No	Yes

Companies - Companies (principals)

Column	Mandatory	What it means	Allowed values	Example
Company Code	Yes	Existing company code / marking from the ERP		ABC
Company Name	Yes			ABC Pharma Ltd
Recognition Code		Short code used in the Invoice Recognition Code (blank = the company code)		ABC
Source ERP		ERP Code from the ERP_Systems sheet / system		SAP
Relationship			PRINCIPAL, OWN	PRINCIPAL
GSTIN				
Contact Person				
Contact Number				
Email				
Address				
Active			Yes / No	Yes

Depots - Depots (godowns / dispatch locations)

Column	Mandatory	What it means	Allowed values	Example
Depot Code	Yes	Unique, also within the same city		GUMA
Depot Name	Yes			Guma Depot
Locality / Area				Guma
City	Yes			Raipur
State				Chhattisgarh
PIN Code				493111
Address				Plot 7, Guma
Recognition Code		Blank = the depot code		GUMA
Companies Handled		Company codes separated by commas	codes, comma separated	ABC, XYZ
Source ERP				MARG
GSTIN				
Contact Person				
Contact Number				
Active			Yes / No	Yes

Customer_Types - Customer types

Column	Mandatory	What it means	Allowed values	Example
Type Key	Yes	Used in the Customers sheet		RETAILER
Type Name	Yes			Retailer
Recognition Code		Short code for the Invoice Recognition Code		RET
Description				
Active			Yes / No	Yes

Customers - Customers

Column	Mandatory	What it means	Allowed values	Example
Customer Code	Yes	Central code		C025
Customer Name	Yes			Sharma Medical Store
Customer Type	Yes	Type Key or Type Name from Customer_Types		RETAILER
Recognition Code		Blank = the customer code		C025
Billing Address				
Delivery Address				
City				Raipur
State				Chhattisgarh
PIN Code				
Delivery Location	Yes	Location Code or Location Name		RPR-CITY
Companies		Company codes, comma separated	codes, comma separated	ABC
Depots		Depot codes, comma separated	codes, comma separated	GUMA
Source ERP				MARG
Source ERP Customer Code		What that ERP calls the customer		SH025
Contact Person				
Mobile				
WhatsApp				
Email				
GSTIN				
Drug Licence No				
Active			Yes / No	Yes

Transporters - Transporters (long-distance and local autos)

Column	Mandatory	What it means	Allowed values	Example
Transporter Code	Yes			SRL
Transporter Name	Yes			Shree Roadlines
Transporter Type	Yes		LONG_DISTANCE, LOCAL_AUTO	LONG_DISTANCE
Payment Mode		Long-distance: TBB = bills later (default) · CASH = paid at dispatch	TBB, CASH	TBB
Charging Method		Local auto only	PER_TRIP, PER_BOX, BOTH	PER_TRIP
Usual Method		Local auto with BOTH	PER_TRIP, PER_BOX	
Usual Trip Charge		Local auto per trip: only a suggestion	number	500
Default Rate per Box		Local auto per box	number	20
Vehicle No				
Contact Person				
Mobile				
GSTIN				
PAN				
City				
State				
Address				
Areas Served				
Direct Pickup Charge		Long-distance: extra when they collect from our depot	number	
Direct Pickup Basis			PER_TRIP, PER_BOX	
Active			Yes / No	Yes

Local_Auto_Rates - Local auto box rates

Column	Mandatory	What it means	Allowed values	Example
Rate Code	Yes			LA-B-RPR
Local Auto Code	Yes	Transporter Code of a LOCAL_AUTO		LA-B
Location		Location Code or Name (blank = all)		
Company		Company Code (blank = all)		
Depot		Depot Code (blank = all)		
Rate per Box	Yes		number	20
Effective From	Yes		DD-MM-YYYY	01-10-2026
Effective To			DD-MM-YYYY	
Remarks				

Freight_Rates - Transporter freight rates (rate cards)

Column	Mandatory	What it means	Allowed values	Example
Transporter Code	Yes			SRL
Location	Yes	Location Code or Name		BSP-CITY
Rate per Case	Yes		number	22
Effective From	Yes		DD-MM-YYYY	01-10-2026

Invoice_Code_Mappings - Invoice code mappings (existing ERP invoice markings)

Column	Mandatory	What it means	Allowed values	Example
Mapping Code	Yes			MARG-XYZ
Source ERP		Blank = any ERP		MARG
Marking	Yes	e.g. XY/ at the start of the ERP invoice number		XY/
Where			PREFIX, CONTAINS	PREFIX
Company	Yes			XYZ
Depot				
Customer Type				
Notes				

ERP_Mappings - ERP code mappings

Column	Mandatory	What it means	Allowed values	Example
Source ERP	Yes			CBO
Entity	Yes		CUSTOMER, COMPANY, DEPOT, LOCATION, TRANSPORTER	CUSTOMER
ERP Code	Yes			CB-0025
Name in ERP				
Central Code	Yes	Code of the record in this system		C025

9. Problems you may meet, and what to do

What you see	Why	What to do
Workbook row 'rejected - ... is mandatory'	A mandatory column is empty	Fill it, upload the corrected rows again
'... not found (not in this workbook and not in CTDMS)'	A link points to a code that does not exist (e.g. delivery location)	Add the missing record or correct the code
'already exists in CTDMS - skipped'	The code is already in the system	Nothing - edit it in Master Settings if it changed
'appears more than once in this sheet'	Same code twice in the workbook	Keep one row
Invoice shows 'code not generated'	A company / depot / customer / customer type has no recognition code	Fill the code in Master Settings; codes are re-made automatically (or press Regenerate)
Import 'Depot does not handle company ...'	The depot's 'Companies handled' does not include the company	Tick the company on the depot
Gatepass 'Enter the trip charge'	Per-trip local auto without an amount	Type the actual trip amount
Gatepass 'no per-box rate for ...'	Per-box local auto has no rate for that location / company	Add a Local Auto Box Rate, or use another auto
'... is charged per trip only'	The chosen method is not allowed for that auto	Choose the allowed method or change the auto's settings
Long-distance 'no rate'	Rate card has no rate for the area (as in v1.5)	Add the rate; the checker approves it
A function key shows 'not used on this screen'	That key has no job on this screen	Look at the key bar at the bottom, or press F1
Esc asks 'Leave this screen?'	Something was typed and not saved	N (or Enter) stays; Y leaves without saving
'Generate gatepass' is grey when Enter reaches it	Something is still missing	The message says what (e.g. choose the transporter); Shift+Enter goes back
A choice box says 'Nothing matches'	The typed code or name is not in that master (or is inactive)	Type less, check the code in Master Settings, or Esc to keep the old value

10. Before real use

1. Start from an empty database (init-db, create-admin) - do not reuse the demonstration database.
2. Prepare the real master workbook from MARG, SAP, CBO, Intrect and other ERPs; upload, check the preview, import.
3. Set the recognition codes you want to use, or switch the Invoice Recognition Code off for the pilot.
4. Enter local autos with their charging method; add per-box rates where needed.
5. Create the real users and roles; keep paid services locked unless the business decides otherwise.
6. Back up the database and test a restore.

Audit entries written during the demonstration: 129.